14. Interrupt Controller Unit (ICU)

14.1 Overview

The Interrupt Controller Unit (ICU) controls which event signals are linked to the Nested Vector Interrupt Controller (NVIC), DMA Controller (DMAC) module and the Data Transfer Controller (DTC) modules. The ICU also controls non-maskable interrupts.

Table 14.1 lists the ICU specifications, Figure 14.1 shows a block diagram, and Table 14.2 lists the I/O pins.

Table 14.1 ICU specifications (1 of 2)

Parameter		Description
Maskable interrupts	Peripheral function interrupts	Interrupts from peripheral modules Number of sources: 497 (a factor is chosen with an event list number 33 to 1023)
	External pin interrupts	● Interrupt detection on low level*4, falling edge, rising edge, rising and falling edges. One of these detection methods can be set for each source ● Digital filter function supported ● 32 sources, with interrupts from IRQi (i = 0 to 31) pins.
	CPU mutual interrupt	● IPC CPU mutual interrupt 0 (IPC_IRQ0) ● IPC CPU mutual interrupt 1 (IPC_IRQ1)
	Interrupt requests to CPU (NVIC)	96 interrupt requests are output to NVIC.
	DMAC control	● The DMAC can be activated using interrupt sources*1 ● The target interrupt source can be selected individually for every DMAC channels.
	DTC control	● The DTC can be activated using interrupt sources*1 ● The method for selecting an interrupt source is the same as that of the interrupt request to NVIC.
Non-maskable interrupts*2	NMI pin interrupt	● Interrupt from the NMI pin ● Interrupt detection on falling edge or rising edge ● Digital filter function supported
	Oscillation stop detection interrupt*3	Interrupt on detection of main oscillation is stopped
	Sub Oscillation stop detection interrupt*3	Interrupt on detection of sub oscillation is stopped
	WDT underflow/refresh error*3	Interrupt on an underflow of the down-counter or occurrence of a refresh error (CPU0, CPU1)
	IWDT underflow/refresh error*3	Interrupt on an underflow of the down-counter or occurrence of a refresh error
	Voltage-monitor 1 interrupt*3	Voltage monitor interrupt of the voltage detection circuit 1 (PVD_PVD1)
	Voltage-monitor 2 interrupt*3	Voltage monitor interrupt of the voltage detection circuit 2 (PVD_PVD2)
	Common memory error interrupt	Common memory error means SRAM ECC error.
	Local memory error interrupt	Local memory error interrupt includes Cache and TCM ECC Error. (CPU1 only)
	Bus error Interrupt	Bus error interrupt includes MPU and TZF error
	Lockup error interrupt	Lockup error interrupt (CPU0, CPU1)
	CPU mutual Interrupt	IPC NMI CPU mutual interrupt (CPU0, CPU1)
	FPU Exception interrupt	FPU Exception interrupt (CPU0, CPU1)
	MRAM read error interrupt	● MRAM read error interrupt for MRC ● MRAM read error interrupt for MRE
Security	Secure	Some registers have Security Attribution.
	Privilege	Each register of the ICU can only be accessed with Privilege access.

Table 14.1 ICU specifications (2 of 2)

Parameter	Description
Return from low power mode	● CPU Sleep mode: Return is initiated by non-maskable interrupts or any other interrupt source. ● CPU Deep Sleep mode: Return is initiated by non-maskable interrupts. Interrupt can be selected as WUPEN0, WUPEN1, DSLPWUPIRQENj register (j = 0 to 2). See section 14.2.4. ICUSARF : Interrupt Controller Unit Security Attribution Register F, section 14.2.9. ICUSARK : Interrupt Controller Unit Security Attribution Register K, section 14.2.10. ICUSARL : Interrupt Controller Unit Security Attribution Register L, section 14.2.11. TEVTRCR : Trusted Event Route Control Register, section 14.9.2. Return from CPU Deep Sleep Mode. ● Software Standby mode: Return is initiated by non-maskable interrupts. Interrupt can be selected as WUPEN0, WUPEN1 register. See section 14.2.4. ICUSARF : Interrupt Controller Unit Security Attribution Register F, section 14.2.9. ICUSARK : Interrupt Controller Unit Security Attribution Register K, section 14.2.10. ICUSARL : Interrupt Controller Unit Security Attribution Register L, section 14.9.2. Return from CPU Deep Sleep Mode.

Note 1. For the DMAC and DTC activation sources, see Table 14.4.
Note 2. Non-maskable interrupts can be enabled only once after a reset release.
Note 3. These non-maskable interrupts can also be used as interrupts in general. When used as maskable interrupts, do not change the value of the NMIER register from the reset state. To enable voltage monitor 1 to 2 interrupts, set the PVD1CR1.PVD1IRQSEL and PVD2CR1.PVD2IRQSEL bits to 1.
Note 4. Low level: interrupt detection is not canceled if you do not clear it after a detection.

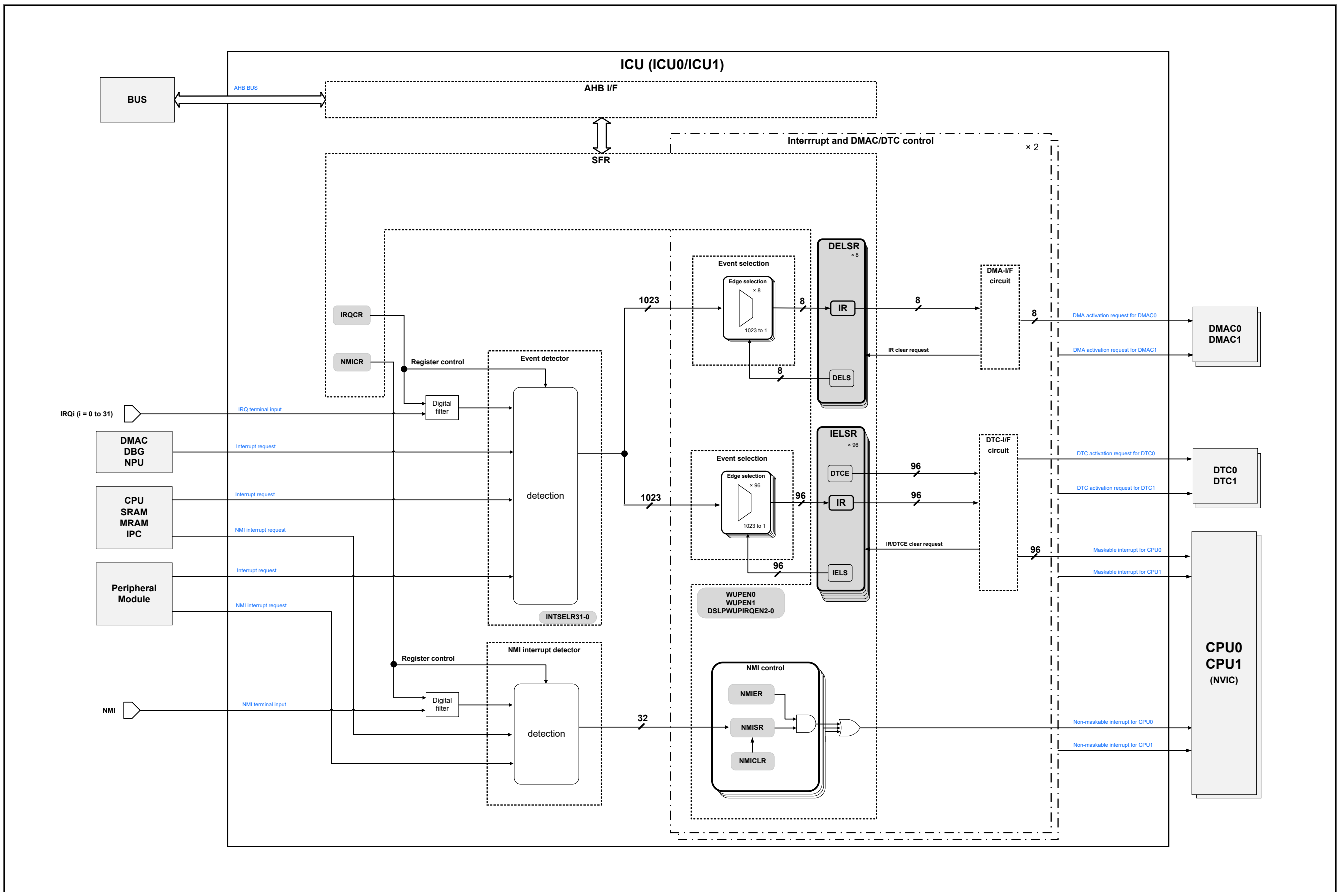

Figure 14.1 ICU block diagram

NMISR: Non-Maskable Interrupt Status Register

NMIER: Non-Maskable Interrupt Enable Register

NMICLR: Non-Maskable Interrupt Status Clear Register

NMICR: NMI Pin Interrupt Control Register

IRQCRi: IRQi Control Register (i = 0 to 31)

WUPEN0/1: Wake Up Interrupt Enable Register0/1

IELSRn: ICU Event Link Setting Register (n = 0 to 95)

DELSRm: DMAC Event Link Setting Register (m = 0 to 7)

IR: Interrupt Status Flag (IELSRn.IR, DELSRm.IR)

DTCE: DTC Activation Enable (IELSRn.DTCE)

DSLPWUPIRQENj: Deep Sleep Wake Up IRQ Enable Register (j = 0 to 2)

INTSELRp: Interrupt request select Register (p = 0 to 31)

Table 14.2 ICU I/O pins

Pin name	I/O	Description
NMI	Input	Non-maskable interrupt request pin
IRQi (i = 0 to IRQ31)	Input	External interrupt request pins

14.2 Register Descriptions

This section does not describe registers internal to Arm® NVIC.

For information on these registers, see the NVIC chapter of Arm® Processor Technical Reference Manual [1], [3].

ICU_BASE (base address of ICU) is 0x4000_C000(Secure) and 0x5000_C000(Non-secure)

COMMON_ICU_BASE (base address of COMMON_ICU) is 0x4000_6000(Secure) and 0x5000_6000(Non-secure)

CPSCU_BASE (base address of CPSCU) is 0x4000_8000(secure) and 0x5000_8000(non-secure).

The ICU0 and ICU1 registers have the same base address. ICU0 is dedicated to CPU0 and can only be accessed from CPU0. ICU1 is dedicated to CPU1 and can only be accessed from CPU1.

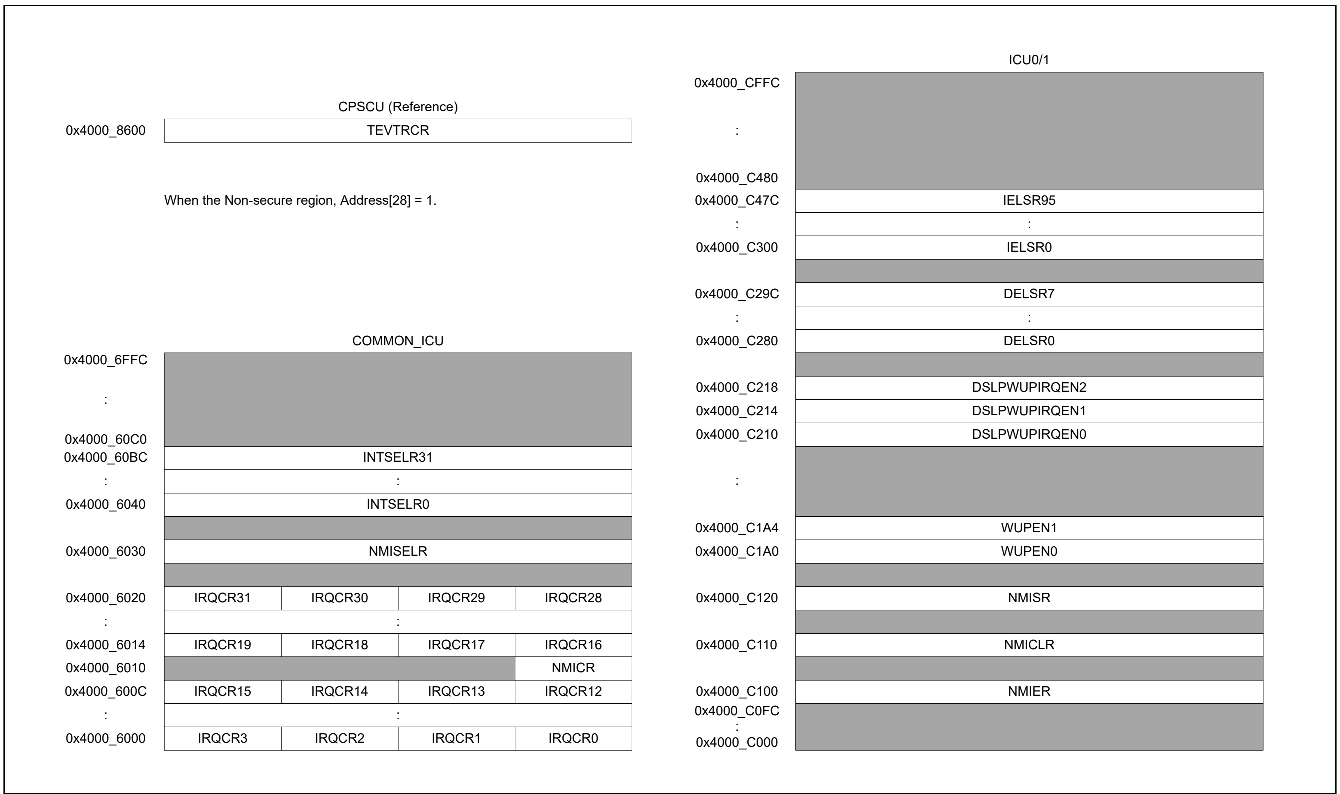

Figure 14.2 ICU Register Address Map

14.2.1 ICUSARA : Interrupt Controller Unit Security Attribution Register A

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x40

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIRQCR31	SAIRQCR30	SAIRQCR29	SAIRQCR28	SAIRQCR27	SAIRQCR26	SAIRQCR25	SAIRQCR24	SAIRQCR23	SAIRQCR22	SAIRQCR21	SAIRQCR20	SAIRQCR19	SAIRQCR18	SAIRQCR17	SAIRQCR16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAIRQCR15	SAIRQCR14	SAIRQCR13	SAIRQCR12	SAIRQCR11	SAIRQCR10	SAIRQCR9	SAIRQCR8	SAIRQCR7	SAIRQCR6	SAIRQCR5	SAIRQCR4	SAIRQCR3	SAIRQCR2	SAIRQCR1	SAIRQCR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SAIRQCR31 to SAIRQCR0	Security attributes of registers for the IRQCR, WUPEN0, WUPEN1 registers 0: Secure 1: Non-Secure	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4.

SAIRQCRn (n = 0 to 31)

Security attributes of registers for IRQCRn, WUPEN0, WUPEN1 registers. The target registers are as follows:

- ICU.IRQCR0 to ICU.IRQCR31 registers
- ICU0.WUPEN0[15:0]
- ICU0.WUPEN1[31:16]
- ICU1.WUPEN0[15:0]
- ICU1.WUPEN1[31:16]

14.2.2 ICUSARB : Interrupt Controller Unit Security Attribution Register B

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x44

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SANMI1	SANMI0	SANMI
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SANMI	Security attributes of the NMICR register 0: Secure 1: Non-Secure	R/W
1	SANMI0	Security attributes of the ICU0.NMISR, ICU0.NMIER, ICU0.NMICLR registers 0: Secure 1: Non-Secure	R/W

Bit	Symbol	Function	R/W
2	SANMI1	Security attributes of the ICU1.NMISR, ICU1.NMIER, ICU1.NMICLR registers 0: Secure 1: Non-Secure	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4.

SANMI bit (Security attributes of the NMICR register)

Security attributes of registers for non-maskable interrupt. The target register is as follows:

- NMICR

SANMI0 bit (Security attributes of the ICU0.NMISR, ICU0.NMIER, ICU0.NMICLR registers)

Security attributes of registers for non-maskable interrupt. The target registers are as follows:

- ICU0. NMIER
- ICU0. NMICLR
- ICU0. NMISR

SANMI1 bit (Security attributes of the ICU1.NMISR, ICU1.NMIER, ICU1.NMICLR registers)

Security attributes of registers for non-maskable interrupt. The target registers are as follows:

- ICU1. NMIER
- ICU1. NMICLR
- ICU1. NMISR

The AIRCR.BFHFNMINS (bit 13) value in Application Interrupt and Reset Control Register of ARM CPU should be the same as the value of security attribution. The initial values of AIRCR.BFHFNMINS and the SANMI bit are different.

AIRCR.BFHFNMINS is secure and SANMI is non-secure. Polarity has the same meaning. Program these to match.

Note: Only one of Secure and Non-Secure can set security attribution for non-maskable interrupt related registers. If you programmed the Secure attribute as secure, it will always jump to the Secure interrupt handler. If you want to release any of the non-maskable interrupt sources to the non-secure user, prepare a function for executing a nonsecure program from the interrupt handler for Secure.

14.2.3 ICUSARE : Interrupt Controller Unit Security Attribution Register E

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x50

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIIC0WUP	SAAGT1CBWUP	SAAGT1CAWUP	SAAGT1UDWUP	SAUSBFS0WUP	SAUSBHSWUP	SARTCPRDWUP	SARTCALMWUP	—	—	—	SAVBATTWUP	SAPVD2WUP	SAPVD1WUP	—	SAIWDTWUP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
16	SAIWDTWUP	Security attributes of registers for WUPEN0.b16 0: Secure 1: Non-Secure	R/W
17	—	This bit is read as 0. The write value should be 0.	R/W
18	SAPVD1WUP	Security attributes of registers for WUPEN0.b18 0: Secure 1: Non-Secure	R/W
19	SAPVD2WUP	Security attributes of registers for WUPEN0.b19 0: Secure 1: Non-Secure	R/W
20	SAVBATTWUP	Security attributes of registers for WUPEN0.b20 0: Secure 1: Non-Secure	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
24	SARTCALMWUP	Security attributes of registers for WUPEN0.b24 0: Secure 1: Non-Secure	R/W
25	SARTCPRDWUP	Security attributes of registers for WUPEN0.b25 0: Secure 1: Non-Secure	R/W
26	SAUSBHSWUP	Security attributes of registers for WUPEN0.b26 0: Secure 1: Non-Secure	R/W
27	SAUSBFS0WUP	Security attributes of registers for WUPEN0.b27 0: Secure 1: Non-Secure	R/W
28	SAAGT1UDWUP	Security attributes of registers for WUPEN0.b28 0: Secure 1: Non-Secure	R/W
29	SAAGT1CAWUP	Security attributes of registers for WUPEN0.b29 0: Secure 1: Non-Secure	R/W
30	SAAGT1CBWUP	Security attributes of registers for WUPEN0.b30 0: Secure 1: Non-Secure	R/W
31	SAIIC0WUP	Security attributes of registers for WUPEN0.b31 0: Secure 1: Non-Secure	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4.

14.2.4 ICUSARF : Interrupt Controller Unit Security Attribution Register F

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x54

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAPDMWUP	SAULP1BWUP	SAULP1AWUP	SAULP1UWUP	SAI3CWUP	SAULP0BWUP	SAULP0AWUP	SAULP0UWUP	SASOSCWUP	—	—	—	SACOMPHS0WUP	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	SACOMPHS0WUP	Security attributes of registers for WUPEN1.b3 0: Secure 1: Non-Secure	R/W
6:4	—	These bits are read as 0. The write value should be 0.	R/W
7	SASOSCWUP	Security attributes of registers for WUPEN1.b7 0: Secure 1: Non-Secure	R/W
8	SAULP0UWUP	Security attributes of registers for WUPEN1.b8 0: Secure 1: Non-Secure	R/W
9	SAULP0AWUP	Security attributes of registers for WUPEN1.b9 0: Secure 1: Non-Secure	R/W
10	SAULP0BWUP	Security attributes of registers for WUPEN1.b10 0: Secure 1: Non-Secure	R/W
11	SAI3CWUP	Security attributes of registers for WUPEN1.b11 0: Secure 1: Non-Secure	R/W
12	SAULP1UWUP	Security attributes of registers for WUPEN1.b12 0: Secure 1: Non-Secure	R/W
13	SAULP1AWUP	Security attributes of registers for WUPEN1.b13 0: Secure 1: Non-Secure	R/W
14	SAULP1BWUP	Security attributes of registers for WUPEN1.b14 0: Secure 1: Non-Secure	R/W
15	SAPDMWUP	Security attributes of registers for WUPEN1.b15 0: Secure 1: Non-Secure	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4.

14.2.5 ICUSARG : Interrupt Controller Unit Security Attribution Register G

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x70

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIELSR31	SAIELSR30	SAIELSR29	SAIELSR28	SAIELSR27	SAIELSR26	SAIELSR25	SAIELSR24	SAIELSR23	SAIELSR22	SAIELSR21	SAIELSR20	SAIELSR19	SAIELSR18	SAIELSR17	SAIELSR16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAIELSR15	SAIELSR14	SAIELSR13	SAIELSR12	SAIELSR11	SAIELSR10	SAIELSR9	SAIELSR8	SAIELSR7	SAIELSR6	SAIELSR5	SAIELSR4	SAIELSR3	SAIELSR2	SAIELSR1	SAIELSR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SAIELSR31 to SAIELSR0	Security attributes of registers for ICU0 event link setting0 0: Secure 1: Non-Secure	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4.

SAIELSRn (n = 0 to 31)

Security attributes of registers for ICU0 event link setting0. The target registers are as follows:

- ICU0.IELSR0 to IELSR31
- ICU0.DSLPWUPIRQEN0

The Secure Attribute managed within the Arm® CPU NVIC must match the security attribution of ICU0.IELSEn (n = 0 to 31). NVIC internal registers are in NVIC_ITNS0[31:0]. Polarity has the same meaning. Program these to match.

14.2.6 ICUSARH : Interrupt Controller Unit Security Attribution Register H

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x74

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIELSR63	SAIELSR62	SAIELSR61	SAIELSR60	SAIELSR59	SAIELSR58	SAIELSR57	SAIELSR56	SAIELSR55	SAIELSR54	SAIELSR53	SAIELSR52	SAIELSR51	SAIELSR50	SAIELSR49	SAIELSR48
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAIELSR47	SAIELSR46	SAIELSR45	SAIELSR44	SAIELSR43	SAIELSR42	SAIELSR41	SAIELSR40	SAIELSR39	SAIELSR38	SAIELSR37	SAIELSR36	SAIELSR35	SAIELSR34	SAIELSR33	SAIELSR32
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SAIELSR32 to SAIELSR63	Security attributes of registers for ICU0event link setting1. 0: Secure 1: Non-Secure	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4.

SAIELSRn (n = 32 to 63)

Security attributes of registers for ICU0 event link setting1. The target registers are as follows:

- ICU0.IELSR32 to IELSR63

- ICU0.DSLPWUPIRQEN1

The Secure Attribute managed within the Arm® CPU NVIC must match the security attribution of ICU0.IELSRn (n = 32 to 63). NVIC internal registers are in NVIC_ITNS1[31:0]. Polarity has the same meaning. Program these to match.

14.2.7 ICUSARI : Interrupt Controller Unit Security Attribution Register I

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x78

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIELSR95	SAIELSR94	SAIELSR93	SAIELSR92	SAIELSR91	SAIELSR90	SAIELSR89	SAIELSR88	SAIELSR87	SAIELSR86	SAIELSR85	SAIELSR84	SAIELSR83	SAIELSR82	SAIELSR81	SAIELSR80
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAIELSR79	SAIELSR78	SAIELSR77	SAIELSR76	SAIELSR75	SAIELSR74	SAIELSR73	SAIELSR72	SAIELSR71	SAIELSR70	SAIELSR69	SAIELSR68	SAIELSR67	SAIELSR66	SAIELSR65	SAIELSR64
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SAIELSR64 to SAIELSR95	Security attributes of registers for ICU0 event link setting2. 0: Secure 1: Non-Secure	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4.

SAIELSRn (n = 64 to 95)

Security attributes of registers for ICU0 event link setting2. The target registers are as follows:

- ICU0.IELSR64 to IELSR95
- ICU0.DSLPWUPIRQEN2

The Secure Attribute managed within the Arm® CPU NVIC must match the security attribution of ICU0.IELSRn (n = 64 to 95). NVIC internal registers are in NVIC_ITNS2[31:0]. Polarity has the same meaning. Program these to match.

14.2.8 ICUSARJ : Interrupt Controller Unit Security Attribution Register J

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x7C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIELSR31	SAIELSR30	SAIELSR29	SAIELSR28	SAIELSR27	SAIELSR26	SAIELSR25	SAIELSR24	SAIELSR23	SAIELSR22	SAIELSR21	SAIELSR20	SAIELSR19	SAIELSR18	SAIELSR17	SAIELSR16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAIELSR15	SAIELSR14	SAIELSR13	SAIELSR12	SAIELSR11	SAIELSR10	SAIELSR9	SAIELSR8	SAIELSR7	SAIELSR6	SAIELSR5	SAIELSR4	SAIELSR3	SAIELSR2	SAIELSR1	SAIELSR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SAIELSR0 to SAIELSR31	Security attributes of registers for ICU1 event link setting0. 0: Secure 1: Non-Secure	R/W

Note: S-TYPE1, P-TYPE1

Note: This register is write-protected by PRCR_S.PRC4.

SAIELSRn (n = 0 to 31)

Security attributes of registers for ICU1 event link setting0. The target registers are as follows:

- ICU1.IELSR0 to IELSR31
- ICU1.DSLPWUPIRQEN0

The Secure Attribute managed within the Arm® CPU NVIC must match the security attribution of ICU1.IELSRn (n = 0 to 31). NVIC internal registers are in NVIC_ITNS0[31:0]. Polarity has the same meaning. Program these to match.

14.2.9 ICUSARK : Interrupt Controller Unit Security Attribution Register K

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x80

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIELSR63	SAIELSR62	SAIELSR61	SAIELSR60	SAIELSR59	SAIELSR58	SAIELSR57	SAIELSR56	SAIELSR55	SAIELSR54	SAIELSR53	SAIELSR52	SAIELSR51	SAIELSR50	SAIELSR49	SAIELSR48
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAIELSR47	SAIELSR46	SAIELSR45	SAIELSR44	SAIELSR43	SAIELSR42	SAIELSR41	SAIELSR40	SAIELSR39	SAIELSR38	SAIELSR37	SAIELSR36	SAIELSR35	SAIELSR34	SAIELSR33	SAIELSR32
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SAIELSR32 to SAIELSR63	Security attributes of registers for ICU1 event link setting1 0: Secure 1: Non-Secure	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4.

SAIELSR (n = 32 to 63)

Security attributes of registers for ICU1 event link setting1. The target registers are as follows:

- ICU1.IELSR32 to IELSR63
- ICU1.DSLPWUPIRQEN1

The Secure Attribute managed within the Arm® CPU NVIC must match the security attribution of ICU1.IELSRn (n = 32 to 63). NVIC internal registers are in NVIC_ITNS1[31:0]. Polarity has the same meaning. Program these to match.

14.2.10 ICUSARL : Interrupt Controller Unit Security Attribution Register L

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x84

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SAIELSR95	SAIELSR94	SAIELSR93	SAIELSR92	SAIELSR91	SAIELSR90	SAIELSR89	SAIELSR88	SAIELSR87	SAIELSR86	SAIELSR85	SAIELSR84	SAIELSR83	SAIELSR82	SAIELSR81	SAIELSR80
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SAIELSR79	SAIELSR78	SAIELSR77	SAIELSR76	SAIELSR75	SAIELSR74	SAIELSR73	SAIELSR72	SAIELSR71	SAIELSR70	SAIELSR69	SAIELSR68	SAIELSR67	SAIELSR66	SAIELSR65	SAIELSR64
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	SAIELSR64 to SAIELSR95	Security attributes of registers for ICU1 event link setting2. 0: Secure 1: Non-Secure	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4.

SAIELSR (n = 64 to 95)

Security attributes of registers for ICU1 event link setting2. The target registers are as follows:

- ICU1.IELSR64 to IELSR95
- ICU1.DSLPWUPIRQEN2

The Secure Attribute managed within the ARM CPU NVIC must match the security attribution of ICU1.IELSRn (n = 64 to 95). NVIC internal registers are in NVIC_ITNS2[31:0]. Polarity has the same meaning. Program these to match.

14.2.11 TEVTRCR : Trusted Event Route Control Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x600

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	TEVTEICU1	TEVTEICU0	TEVTE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TEVTE	Trusted Event Route Control Register for ELC 0: Disable 1: Enable	R/W
1	TEVTEICU0	Trusted Event Route Control Register for ICU0 0: Disable 1: Enable	R/W
2	TEVTEICU1	Trusted Event Route Control Register for ICU1 0: Disable 1: Enable	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE1, P-TYPE1
Note: This register is write-protected by PRCR_S.PRC4

TEVTE bit (Trusted Event Route Control Register for ELC)

When TEVTE = 1, all ELS [9:0] bits of ELC.ELSRn (n = 0 to18) are allowed secure access write. Non-secure access write is protected.

- ELSRn.IELS[9:0] (n = 0 to 18)

TEVTEICU0 bit (Trusted Event Route Control Register for ICU0)

When TEVTEICU0 = 1, the IELS [9:0] bits of ICU0.IELSRn (n = 0 to 95), all DELS [9:0] bits of ICU0.DELSRm (m = 0 to 7) are allowed secure access write. Non-secure access write is protected. Additionally, when TEVTEICU0 = 1, if the Security Attribution of the target register (IELS [31:16] bits of ICU0.IELSRn (n = 0 to 95)) is non-secure, then secure access is not allowed. At this time, the upper level [31:16] cannot be read and write, but the response is OK, and no error occurs.

- ICU0.IELSRn.IELS[9:0] (n = 0 to 95)
- ICU0.DELSRm.DELS[9:0] (m = 0 to 7)

TEVTEICU1 bit (Trusted Event Route Control Register for ICU1)

When TEVTEICU1 = 1, the IELS [9:0] bits of ICU1.IELSRn (n = 0 to 95), the all DELS [9:0] bits of ICU1.DELSRm (m = 0 to 7) are allowed secure access write. Non-secure access write is protected. Additionally, when TEVTEICU1 = 1, if the Security Attribution of the target register (IELS [31:16] bits of ICU1.IELSRn (n = 0 to 95)) is non-secure, then secure access is not allowed. At this time, the upper level [31:16] cannot be read and write, but the response is OK, and no error occurs.

- ICU1.IELSRn.IELS[9:0] (n = 0 to 95)
- ICU1.DELSRm.DELS[9:0] (m = 0 to 7)

14.2.12 IRQCRi : IRQ Control Register (i = 0 to 31)

Base address: ICU_COMMON = 0x4000_6000
ICU_COMMON_NS = 0x5000_6000

Offset address: 0x000 + 0x01 × i (i = 0 to 15)
0x004 + 0x01 × i (i = 16 to 31)

Bit position:	7	6	5	4	3	2	1	0
Bit field:	FLTEN	—	FCLKSEL[1:0]		—	—	IRQMD[1:0]	
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	IRQMD[1:0]	IRQi Detection Sense Select 0 0: Falling edge 0 1: Rising edge 1 0: Rising and falling edges 1 1: Low level	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
5:4	FCLKSEL[1:0]	IRQi Digital Filter Sampling Clock Select 0 0: PCLKB 0 1: PCLKB/8 1 0: PCLKB/32 1 1: PCLKB/64	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	FLTEN	IRQi Digital Filter Enable 0: Digital filter is disabled 1: Digital filter is enabled.	R/W

Note: S-TYPE3, P-TYPE2

IRQCRi register changes must satisfy the following conditions:

1. For a CPU interrupt or DTC trigger:
   Change the IRQCRi register value before setting the target IELSRn register (n = 0 to 95).
   The register value should be changed only when the value of the target IELSRn register is 0x0000.
2. For a DMAC trigger:
   Change the IRQCRi register value before setting the target DELSRm register (m = 0 to 7).
   The register value should be changed only when the value of the target DELSRm register is 0x0000.
3. For a wakeup enable signal:
   Change the IRQCRi register setting before setting the target WUPEN0.IRQWUPEN[n] (n = 0 to 15) or WUPEN1.IRQWUPEN[m] (m = 31 to 16). You can only change the register values when the target WUPEN0.IRQWUPEN[n] or WUPEN1.IRQWUPEN[m] is 0.

IRQMD[1:0] bits (IRQi Detection Sense Select)

The IRQMD[1:0] bits set the detection sensing method for the IRQi external pin interrupt sources. For setting method when using external pin interrupt, see section 14.5.7. External Pin Interrupts.

FCLKSEL[1:0] bits (IRQi Digital Filter Sampling Clock Select)

The FCLKSEL[1:0] bits select the digital filter sampling clock for the IRQi external pin interrupt request pins, selectable to:

- PCLKB (every cycle)
- PCLKB/8 (once every 8 cycles)
- PCLKB/32 (once every 32 cycles)
- PCLKB/64 (once every 64 cycles)

For details of the digital filter, see section 14.5.6. Digital Filter.

FLTEN bit (IRQi Digital Filter Enable)

The FLTEN bit enables the digital filter used for the IRQi external pin interrupt sources. The digital filter is enabled when the IRQCRi.FLTEN bit is 1 and disabled when the IRQCRi.FLTEN bit is 0. The IRQi pin level is sampled at the clock cycle specified in the IRQCRi.FCLKSEL[1:0] bits. When the sampled level matches three times, the output level from the digital filter changes. For details of the digital filter, see section 14.5.6. Digital Filter.

14.2.13 NMISR : Non-Maskable Interrupt Status Register

Base address: ICU = 0x4000_C000
ICU_NS = 0x5000_C000

Offset address: 0x120

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	IPCST	—	MRERDST	MRCRDST	FPUEXCST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LUST	LMST	CMST	BUSST	—	—	—	—	NMIST	OSTST	SOSTST	—	PVD2ST	PVD1ST	WDTST	IWDTST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IWDTST	IWDT Underflow/Refresh Error Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
1	WDTST	WDT Underflow/Refresh Error Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
2	PVD1ST	Voltage Monitor 1 Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
3	PVD2ST	Voltage Monitor 2 Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
4	—	These bits are read as 0.	R
5	SOSTST	Sub Oscillation Stop Detection Interrupt Status Flag 0: Interrupt not requested for sub oscillation stop 1: Interrupt requested for sub oscillation stop	R
6	OSTST	Main Clock Oscillation Stop Detection Interrupt Status Flag 0: Interrupt not requested for main clock oscillation stop 1: Interrupt requested for main clock oscillation stop	R

Bit	Symbol	Function	R/W
7	NMIST	NMI Pin Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
11:8	—	These bits are read as 0.	R
12	BUSST	Bus Error Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
13	CMST	Common Memory Error Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
14	LMST	Local Memory Error Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested.	R
15	LUST	LockUp Error Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
16	FPUEXCST	FPU Exception Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
17	MRCRDST	MRAM MRC read Error Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
18	MRERDST	MRAM MRE read Error Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
19	—	This bit is read as 0.	R
20	IPCST	IPC NMI CPU mutual Interrupt Status Flag 0: Interrupt not requested 1: Interrupt requested	R
31:21	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE2

The NMISR register monitors the status of non-maskable interrupt sources. Writes to the NMISR register are ignored.

NMISR is set with a signal that is valid as a non-maskable interrupt supplied to CPU0/CPU1 by the setting judgment of NMIER. For details of the judgment by the NMIER setting, see section 14.8.2. Trusted Interrupt Management.

Before the end of the non-maskable interrupt handler, check that all of the bits in this register are set to 0 to confirm that no other NMI requests are generated during handler processing.

IWDTST flag (IWDT Underflow/Refresh Error Interrupt Status Flag)

The IWDTST flag indicates an IWDT underflow/refresh error interrupt request. It is read-only and cleared by the NMICLR.IWDTCLR bit. The IWDT interrupt is only valid for primary CPU. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the IWDT underflow/refresh error interrupt is generated and this interrupt source is enabled.

[Clearing condition]

When 1 is written to the NMICLR.IWDTCLR bit.

WDTST flag (WDT Underflow/Refresh Error Interrupt Status Flag)

The WDTST flag indicates a WDT underflow/refresh error interrupt request. It is read-only and cleared by the NMICLR.WDTCLR bit. Accepts WDT interrupt for same CPUs. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the WDT underflow/refresh error interrupt is generated.

[Clearing condition]

When 1 is written to the NMICLR.WDTCLR bit.

PVD1ST flag (Voltage Monitor 1 Interrupt Status Flag)

The PVD1ST flag indicates a request for voltage monitor 1 interrupt. It is read-only and cleared by the NMICLR.PVD1CLR bit. The voltage monitor 1 interrupt is valid only when bit 2 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the voltage monitor 1 interrupt is generated, and this interrupt source is enabled.

[Clearing condition]

When 1 is written to the NMICLR.PVD1CLR bit.

PVD2ST flag (Voltage Monitor 2 Interrupt Status Flag)

The PVD2ST flag indicates a request for voltage monitor 2 interrupt. It is read-only and cleared by the NMICLR.PVD2CLR bit. The voltage monitor 2 interrupt is valid only when bit 3 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the voltage monitor 2 interrupt is generated, and this interrupt source is enabled.

[Clearing condition]

When 1 is written to the NMICLR.PVD2CLR bit.

SOSTST flag (Sub Oscillation Stop Detection Interrupt Status Flag)

The SOSTST flag indicates a sub oscillation stop detection interrupt request. It is read-only and cleared by the NMICLR.SOSTCLR bit. The sub oscillation stop detection interrupt is valid only when bit 5 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the sub oscillation stops detection, interrupt is generated.

[Clearing condition]

When 1 is written to the NMICLR.SOSTCLR bit.

OSTST flag (Main Clock Oscillation Stop Detection Interrupt Status Flag)

The OSTST flag indicates an oscillation stop detection interrupt request. It is read-only and cleared by the NMICLR.OSTCLR bit. The oscillation stop detection interrupt is valid only when bit 6 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the oscillation stops detection, interrupt is generated.

[Clearing condition]

When 1 is written to the NMICLR.OSTCLR bit.

NMIST flag (NMI Pin Interrupt Status Flag)

The NMIST flag indicates a NMI pin interrupt request. It is read-only and cleared by the NMICLR.NMICLR bit. The NMI pin interrupt is valid only when bit 7 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When an edge specified by the NMICR.NMIMD bit is input to the NMI pin.

[Clearing condition]

When 1 is written to the NMICLR.NMICLR bit.

BUSST flag (Bus Error Interrupt Status Flag)

The BUSST flag indicates a request for bus error interrupt as a NMI trigger. It is read-only and cleared by the NMICLR.BUSCLR bit. Bus error interrupt includes MPU and TZF error. The bus error interrupt is valid only when bit 12 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the bus error detection interrupt is generated.

[Clearing condition]

When 1 is written to the NMICLR.BUSCLR bit.

Be sure to clear the error status of the request source before clearing. If CPU does not clear the error status of the request source, the NMI status is set again even if this status is cleared, and an NMI request is issued to the CPU. If the CPU returns from the NMI handler, the CPU jumps back to the NMI handler.

In the case of level detection, use the following steps to clear the Status flag.

(1)Negate the level of an input factor.

(2)Perform a peripheral read access once and make sure the level interrupt is cleared.

(3)Clear the status flag by NMICLR.BUSCLR.

CMST flag (Common Memory Error Interrupt Status Flag)

The CMST flag indicates common memory error. It is read-only and cleared by the NMICLR.CMCLR bit.

Common memory error is SRAM ECC error. The common memory error interrupt is valid only when bit 13 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the common memory error detection interrupt is generated.

[Clearing condition]

When 1 is written to the NMICLR.CMCLR bit.

Be sure to clear the error status of the request source before clearing. If CPU does not clear the error status of the request source, the NMI status is set again even if this status is cleared, and an NMI request is issued to the CPU. If the CPU returns from the NMI handler, the CPU jumps back to the NMI handler.

In the case of level detection, use the following steps to clear the Status flag.

(1)Negate the level of an input factor.

(2)Perform a peripheral read access once and make sure the level interrupt is cleared.

(3)Clear the status flag by NMICLR.CMCLR.

LMST flag (Local Memory Error Interrupt Status Flag)

The LMST flag indicates local memory error. It is read-only and cleared by the NMICLR.LMCLR bit.

Local memory error includes CPU1 cache and TCM ECC error. The local memory error interrupt is valid only when bit 14 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the local memory error detection interrupt is generated.

[Clearing condition]

When 1 is written to the NMICLR.LMCLR bit.

Be sure to clear the error status of the request source before clearing. If CPU does not clear the error status of the request source, the NMI status will be set again even if this status is cleared, and an NMI request will be issued to the CPU. If the CPU returns from the NMI handler, the CPU will jump back to the NMI handler.

In the case of level detection, clear of the Status flag should follow the steps below.

(1)Negate the level of an input factor.

(2)Perform a peripheral read access once and make sure the level interrupt is cleared.

(3)Clear the status flag by NMICLR.LUSTCLR.

LUST flag (LockUp Error Interrupt Status Flag)

The LUST flag indicates la lockup. It is read-only and cleared by the NMICLR. LUCLR bit. Accepts Lockup error interrupt for same CPUs. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the local memory error detection interrupt is generated.

[Clearing condition]

When 1 is written to the NMICLR.LMCLR bit.

Be sure to clear the error status of the request source before clearing. If CPU does not clear the error status of the request source, the NMI status will be set again even if this status is cleared, and an NMI request is issued to the CPU. If the CPU returns from the NMI handler, the CPU will jump back to the NMI handler.

In the case of level detection, use the following steps to clear the Status flag.

(1)Negate the level of an input factor.

(2)Perform a peripheral read access once and make sure the level interrupt is cleared.

(3)Clear the status flag by NMICLR.LUCLR.

FPUEXCST flag (FPU Exception Interrupt Status Flag)

The FPUEXCST flag indicates FPU Exception interrupt. It is read-only and cleared by the NMICLR.FPUEXCCLR bit. Accepts FPU Exception interrupt for same CPU. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the FPU Exception detection interrupt is generated

[Clearing condition]

When 1 is written to the NMICLR.FPUEXCCLR bit

Be sure to clear the error status of the request source before clearing. If CPU does not clear the error status of the request source, the NMI status will be set again even if this status is cleared, and an NMI request will be issued to the CPU. If the CPU returns from the NMI handler, the CPU will jump back to the NMI handler.

In the case of level detection, clear of the Status flag should follow the steps below.

(1)Negate the level of an input factor.

(2)Perform a peripheral read access once and make sure the level interrupt is cleared.

(3)Clear the status flag by NMICLR.FPUEXCCLR.

MRCRDST flag (MRAM MRC read Error Interrupt Status Flag)

The MRCRDST flag indicates MRAM MRC read error interrupt. It is read-only and cleared by the NMICLR.MRCRDCLR bit. The MRAM MRC read error interrupt is valid only when bit 17 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the MRAM MRC read error detection interrupt is generated

[Clearing condition]

When 1 is written to the NMICLR.MRCRDSCLR bit

Be sure to clear the error status of the request source before clearing. If CPU does not clear the error status of the request source, the NMI status will be set again even if this status is cleared, and an NMI request will be issued to the CPU. If the CPU returns from the NMI handler, the CPU will jump back to the NMI handler.

In the case of level detection, clear of the Status flag should follow the steps below.

(1)Negate the level of an input factor.

(2)Perform a peripheral read access once and make sure the level interrupt is cleared.

(3)Clear the status flag by NMICLR.MRCRDCLR.

MRERDST flag (MRAM MRE read Error Interrupt Status Flag)

The MRERDST flag indicates MRAM MRE read error interrupt. It is read-only and cleared by the NMICLR.MRERDCLR bit. The MRAM MRE read error interrupt is valid only when bit 18 of the NMIER register is set to "1" and supply to the target CPU is selected. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the MRAM MRE read error detection interrupt is generated

[Clearing condition]

When 1 is written to the NMICLR.MRERDCLR bit

Be sure to clear the error status of the request source before clearing. If CPU does not clear the error status of the request source, the NMI status will be set again even if this status is cleared, and an NMI request will be issued to the CPU. If the CPU returns from the NMI handler, the CPU will jump back to the NMI handler.

In the case of level detection, clear of the Status flag should follow the steps below.

(1)Negate the level of an input factor.

(2)Perform a peripheral read access once and make sure the level interrupt is cleared.

(3)Clear the status flag by NMICLR.MRERDCLR.

IPCST flag (IPC NMI CPU mutual Interrupt Status Flag)

The IPCST flag indicates a request for IPC NMI CPU mutual interrupt as a NMI trigger. It is read-only and cleared by the NMICLR.IPCCLR bit. Accepts IPC interrupt for other CPUs. See section 14.8.2. Trusted Interrupt Management.

[Setting condition]

When the IPC NMI CPU mutual interrupt is generated.

[Clearing condition]

When 1 is written to the NMICLR.IPCCLR bit.

Be sure to clear the error status of the request source before clearing. If CPU does not clear the error status of the request source, the NMI status is set again even if this status is cleared, and an NMI request is issued to the CPU. If the CPU returns from the NMI handler, the CPU jumps back to the NMI handler.

In the case of level detection, use the following steps to clear the Status flag.

(1)Negate the level of an input factor.

(2)Perform a peripheral read access once and make sure the level interrupt is cleared.

(3)Clear the status flag by NMICLR.IPCCLR.

14.2.14 NMIER : Non-Maskable Interrupt Enable Register

Base address: ICU = 0x4000_C000
ICU_NS = 0x5000_C000

Offset address: 0x100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	IPCEN	—	MRERDEN	MRCRDEN	FPUEXCEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LUEN	LMEN	CMEN	BUSEN	—	—	—	—	NMIEN	OSTEN	SOSTEN	—	PVD2EN	PVD1EN	WDTEN	IWDTEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IWDTEN	IWDT Underflow/Refresh Error Interrupt Enable 0: Disabled 1: Enabled	R/W*1 *2
1	WDTEN	WDT Underflow/Refresh Error Interrupt Enable 0: Disabled 1: Enabled	R/W*1 *2
2	PVD1EN	Voltage monitor 1 Interrupt Enable 0: Disabled 1: Enabled	R/W*1 *2
3	PVD2EN	Voltage monitor 2 Interrupt Enable 0: Disabled 1: Enabled	R/W*1 *2
4	—	This bit is read as 0. The write value should be 0.	R/W
5	SOSTEN	Sub Oscillation Stop Detection Interrupt Enable 0: Disabled 1: Enabled	R/W
6	OSTEN	Oscillation Stop Detection Interrupt Enable 0: Disabled 1: Enabled	R/W*1 *2
7	NMIEN	NMI Pin Interrupt Enable 0: Disabled 1: Enabled	R/W*1
11:8	—	These bits are read as 0. The write value should be 0.	R/W
12	BUSEN	Bus Error Interrupt Enable 0: Disabled 1: Enabled	R/W*1
13	CMEN	Common Memory Error Interrupt Enable 0: Disabled 1: Enabled	R/W*1
14	LMEN	Local Memory Error Interrupt Enable 0: Disabled 1: Enabled	R/W
15	LUEN	LockUp Error Interrupt Enable 0: Disabled 1: Enabled	R/W*1
16	FPUEXCEN	FPU Exception Interrupt Enable 0: Disabled 1: Enabled	R/W
17	MRCRDEN	MRAM MRC read Error Interrupt Enable 0: Disabled 1: Enabled	R/W
18	MRERDEN	MRAM MRE read Error Interrupt Enable 0: Disabled 1: Enabled	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
20	IPCEN	IPC NMI CPU mutual Interrupt Enable 0: Disabled 1: Enabled	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE2
Note 1. You can write 1 to this bit only once after reset. Subsequent write accesses are invalid. Writing 0 to this bit is invalid.
Note 2. Do not write 1 to this bit when the source is used as an interrupt signal.

For the NMIER bit, set only one of CPU0 and CPU1 to "1". Setting "1" to the NMIER bit of both CPU0 and CPU1 is prohibited.

IWDTEN bit (IWDT Underflow/Refresh Error Interrupt Enable)

The IWDTEN bit enables IWDT underflow/refresh error interrupt as a NMI trigger. This bit is the enable bit to control the use of the IWDT underflow/refresh error interrupt as a Deep Sleep/Software Standby return factor.

WDTEN bit (WDT Underflow/Refresh Error Interrupt Enable)

The WDTEN bit enables WDT underflow/refresh error interrupt as a NMI trigger. This bit is the enable bit to control the use of the WDT underflow/refresh error interrupt as a Deep Sleep return factor.

PVD1EN bit (Voltage monitor 1 Interrupt Enable)

The PVD1EN bit enables voltage monitor 1 interrupt as a NMI trigger.

PVD2EN bit (Voltage monitor 2 Interrupt Enable)

The PVD2EN bit enables voltage monitor 2 interrupt as a NMI trigger. This bit is the enable bit to control the use of the voltage monitor 2 interrupt as a Deep Sleep/Software Standby return factor.

SOSTEN bit (Sub Oscillation Stop Detection Interrupt Enable)

The OSTEN bit enables main clock oscillation stop detection interrupt as a NMI trigger. This bit is the enable bit to control the use of the sub oscillation stop detection interrupt as a Deep Sleep/Software Standby return factor.

OSTEN bit (Oscillation Stop Detection Interrupt Enable)

The OSTEN bit enables main clock oscillation stop detection interrupt as a NMI trigger. This bit is the enable bit to control the use of the main oscillation stop detection interrupt as a Deep Sleep/Software Standby return factor.

NMIEN bit (NMI Pin Interrupt Enable)

The NMIEN bit enables NMI pin interrupt as a NMI trigger. This bit is the enable bit to control the use of the NMI pin interrupt as a Deep Sleep/Software Standby return factor.

BUSEN bit (Bus Error Interrupt Enable)

The BUSEN bit enables bus error interrupt as a NMI trigger. This bit is the enable bit to control the use of the bus error interrupt as a Deep Sleep return factor. Bus error includes MPU and TZF errors.

CMEN bit (Common Memory Error Interrupt Enable)

The CMEN bit enables common memory error interrupt as a NMI trigger. This bit is the enable bit to control the use of the bus error interrupt as a Deep Sleep return factor. Common memory error is SRAM ECC error.

LMEN bit (Local Memory Error Interrupt Enable)

LMEN bit enables local memory error interrupt as a NMI trigger. This bit is the enable bit to control the use of the local memory error interrupt as a Deep Sleep return factor. Local memory error interrupt includes Cache and TCM ECC Error.

LUEN bit (LockUp Error Interrupt Enable)

LUEN bit enables LockUp error interrupt as a NMI trigger.

FPUEXCEN bit (FPU Exception Interrupt Enable)

FPUEXCEN bit enables FPU Exception Interrupt as a NMI trigger.

MRCRDEN bit (MRAM MRC read Error Interrupt Enable)

MRCRDEN bit enables MRAM MRC read error Interrupt as a NMI trigger. This bit is the enable bit to control the use of the MRAM MRC read error interrupt as a Deep Sleep return factor.

MRERDEN bit (MRAM MRE read Error Interrupt Enable)

MRERDEN bit enables MRAM MRE read error Interrupt as a NMI trigger. This bit is the enable bit to control the use of the MRAM MRE read error interrupt as a Deep Sleep return factor.

IPCEN bit (IPC NMI CPU mutual Interrupt Enable)

IPCEN bit enables IPC NMI CPU mutual interrupt as a NMI trigger. This bit is the enable bit to control the use of the IPC NMI CPU mutual interrupt as a Deep Sleep return factor.

14.2.15 NMICLR : Non-Maskable Interrupt Status Clear Register

Base address: ICU = 0x4000_C000
ICU_NS = 0x5000_C000

Offset address: 0x110

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	IPCCLR	—	MRERDCLR	MRCRDCLR	FPUEXCCLR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LUCLR	LMCLR	CMCLR	BUSCLR	—	—	—	—	NMICLR	OSTCLR	SOSTCLR	—	PVD2CLR	PVD1CLR	WDTCLR	IWDTCLR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IWDTCLR	IWDT Underflow/Refresh Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.IWDTST flag	R/W*1
1	WDTCLR	WDT Underflow/Refresh Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.WDTST flag	R/W*1
2	PVD1CLR	Voltage Monitor 1 Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.PVD1ST flag	R/W*1
3	PVD2CLR	Voltage Monitor 2 Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.PVD2ST flag.	R/W*1
4	—	This bit is read as 0. The write value should be 0.	R/W
5	SOSTCLR	Oscillation Stop Detection Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.OSTST flag	R/W*1
6	OSTCLR	Oscillation Stop Detection Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.OSTST flag	R/W*1
7	NMICLR	NMI Pin Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.NMIST flag	R/W*1
11:8	—	These bits are read as 0. The write value should be 0.	R/W
12	BUSCLR	Bus Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.BUSST flag	R/W*1
13	CMCLR	Common Memory Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.CMST flag	R/W*1
14	LMCLR	Local Memory Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.LMST flag	R/W*1
15	LUCLR	LockUp Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.LUST flag	R/W*1
16	FPUEXCCLR	FPU Exception Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.FPUEXCST flag	R/W*1

Bit	Symbol	Function	R/W
17	MRCRDCLR	MRAM MRC read Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.MRCRDST flag	R/W*1
18	MRERDCLR	MRAM MRE read Error Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.MRERDST flag	R/W*1
19	—	This bit is read as 0. The write value should be 0.	R/W
20	IPCCLR	IPC NMI CPU mutual Interrupt Status Flag Clear 0: No effect 1: Clear the NMISR.IPCST flag	R/W*1
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE2
Note: There may be a difference in processing speed between the CPU and ICU, and the CPU may exit the interrupt handler before clearing the NMISR. Then the CPU will accidentally jump to the NMI handler again. To avoid this, be sure to read NMISR before exiting the NMI handler and make sure that NMISR is cleared before exiting the NMI handler.
Note 1. Only write 1 to this bit.

IWDTCLR bit (IWDT Underflow/Refresh Error Interrupt Status Flag Clear)

Writing 1 to the IWDTCLR bit clears the NMISR.IWDTST flag. This bit is read as 0.

WDTCLR bit (WDT Underflow/Refresh Error Interrupt Status Flag Clear)

Writing 1 to the WDTCLR bit clears the NMISR.WDTST flag. This bit is read as 0.

PVD1CLR bit (Voltage Monitor 1 Interrupt Status Flag Clear)

Writing 1 to the PVD1CLR bit clears the NMISR.PVD1ST flag. This bit is read as 0.

PVD2CLR bit (Voltage Monitor 2 Interrupt Status Flag Clear)

Writing 1 to the PVD2CLR bit clears the NMISR.PVD2ST flag. This bit is read as 0.

SOSTCLR bit (Oscillation Stop Detection Interrupt Status Flag Clear)

Writing 1 to the SOSTCLR bit clears the NMISR.SOSTST flag. This bit is read as 0.

OSTCLR bit (Oscillation Stop Detection Interrupt Status Flag Clear)

Writing 1 to the OSTCLR bit clears the NMISR.OSTST flag. This bit is read as 0.

NMICLR bit (NMI Pin Interrupt Status Flag Clear)

Writing 1 to the NMICLR bit clears the NMISR.NMIST flag. This bit is read as 0.

BUSCLR bit (Bus Error Interrupt Status Flag Clear)

Writing 1 to the BUSCLR bit clears the NMISR.BUSST flag. This bit is read as 0.

CMCLR bit (Common Memory Error Interrupt Status Flag Clear)

Writing 1 to the CMCLR bit clears the NMISR.CMST flag. This bit is read as 0.

LMCLR bits (Local Memory Error Interrupt Status Flag Clear)

Writing 1 to the LMCLR bit clears the NMISR.LMST flag. This bit is read as 0.

LUCLR bit (LockUp Error Interrupt Status Flag Clear)

Writing 1 to the LUCLR bit clears the NMISR.LUST flag. This bit is read as 0.

FPUEXCCLR bits (FPU Exception Interrupt Status Flag Clear)

Writing 1 to the FPUEXCCLR bit clears the NMISR.FPUEXCST flag. This bit is read as 0.

MRCRDCLR bits (MRAM MRC read Error Interrupt Status Flag Clear)

Writing 1 to the MRCRDCLR bit clears the NMISR.MRCRDST flag. This bit is read as 0.

MRERDCLR bits (MRAM MRE read Error Interrupt Status Flag Clear)

Writing 1 to the MRERDCLR bit clears the NMISR.MRERDST flag. This bit is read as 0

IPCCLR bits (IPC NMI CPU mutual Interrupt Status Flag Clear)

Writing 1 to the IPCCLR bit clears the NMISR.IPCST flag. This bit is read as 0.

14.2.16 NMICR : NMI Pin Interrupt Control Register

Base address: ICU_COMMON = 0x4000_6000
ICU_COMMON_NS = 0x5000_6000

Offset address: 0x10

Bit position:	7	6	5	4	3	2	1	0
Bit field:	NFLTEN	—	NFCLKSEL[1:0]		—	—	—	NMIMD
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NMIMD	NMI Detection Set 0: Falling edge 1: Rising edge	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
5:4	NFCLKSEL[1:0]	NMI Digital Filter Sampling Clock Select 0 0: PCLKB 0 1: PCLKB/8 1 0: PCLKB/32 1 1: PCLKB/64	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	NFLTEN	NMI Digital Filter Enable 0: Disabled 1: Enabled	R/W

Note: S-TYPE3, P-TYPE2

Change the NMICR register settings before enabling NMI pin interrupts, that is, before setting the NMIER.NMIEN bit to 1.

NMIMD bit (NMI Detection Set)

The NMIMD bit selects the detection sensing method for the NMI pin interrupts.

NFCLKSEL[1:0] bits (NMI Digital Filter Sampling Clock Select)

The NFCLKSEL[1:0] bits select the digital filter sampling clock for the NMI pin interrupts, selectable to:

- PCLKB (every cycle)
- PCLKB/8 (once every 8 cycles)
- PCLKB/32 (once every 32 cycles)
- PCLKB/64 (once every 64 cycles)

For details of the digital filter, see section 14.5.6. Digital Filter.

NFLTEN bit (NMI Digital Filter Enable)

The NFLTEN bit enables the digital filter used for NMI pin interrupts. The filter is enabled when NFLTEN bit is 1, and disabled when the NFLTEN bit is 0. The NMI pin level is sampled at the sampling clock cycle specified in NMIFLTC.NFCLKSEL[1:0]. When the sampled level matches three times, the output level from the digital filter changes. For details of the digital filter, see section 14.5.6. Digital Filter.

14.2.17 IELSRn : Interrupt Controller Unit Event Link Setting Register n (n = 0 to 95)

Base address: ICU = 0x4000_C000
ICU_NS = 0x5000_C000

Offset address: 0x300 + 0x4 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	DTCE	—	—	—	—	—	—	—	IR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	IELS[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	IELS[9:0]	ICU Event Link Select 0x00: Disable interrupts to the associated NVIC or DTC module Others: Event signal number to be linked. For details, see section 14.3.2. Event Number.	R/W*1
15:10	—	These bits are read as 0. The write value should be 0.	R/W
16	IR	Interrupt Status Flag 0: No interrupt request generated. 1: An interrupt request is generated.	R/W*2
23:17	—	These bits are read as 0. The write value should be 0.	R/W
24	DTCE	DTC Activation Enable 0: DTC activation is disabled. 1: DTC activation is enabled.	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2
Note 1. The [15:0] bits in this register can only be accessed in halfword or word access. Byte access is ignored.
Note 2. Writing 1 to the IR flag is prohibited.

This register has different secure access permission depending on the setting of Trusted Event Route Control Register (TEVTRCR).

If the security attribution is configured as secure,

- Secure access is allowed
- Non-secure write access is ignored and non-secure read access is read as 0. TrustZone access error is generated.

If the security attribution is configured as non-secure and the trusted event route is disabled:

- Secure write access is ignored and secure read access is read as 0. TrustZone access error is generated.
- Non-secure access is allowed.

If the security attribution is configured as non-secure and the trusted event route is enabled:

- Secure access to IELS bit is allowed
- Non-secure write access to IELS bit is ignored and non-secure read access is allowed
- Secure write access to other bits is ignored and secure read access to other bits is read as 0
- Non-secure access to other bits is allowed
- TrustZone access error is not generated

The IELSRn register selects the interrupt source used by the NVIC. For details, see Table 14.3. IELSRn, where n = 0 to 95, corresponds to the NVIC interrupt input source numbers 0 to 95.

IELS[9:0] bits (ICU Event Link Select)

The IELS[9:0] bits link an event signal to the associated NVIC or DTC module. For details, see section 14.3. Vector Table. All IELS[9:0] bits must be written at the same time.

[Setting condition]

When value is written to these bits.

[Clearing conditions]

When 0 is written to these bits.

When DTC transfer is stopped by the access error occurs. See section 18, Data Transfer Controller (DTC) unit chapter.

IR flag (Interrupt Status Flag)

The IR status flag indicates an individual interrupt request from the event specified in IELS[9:0].

[Setting condition]

When an interrupt request is received from the associated peripheral module or IRQi pin.

[Clearing condition]

- The IR flag is cleared to 0 by writing 0.
- In the case of DTC.DISEL = 0. At the time other than the final transfer end in DTC transfer during DTCE = 1, IR flag repeat set and cleared by Hardware.
- In the case of DTC.DISEL = 1. For DTC transfers during DTCE = 1, the hardware does not clear the IR flag. Must be cleared by the CPU like No.1.

When DTC transfer except last transfer is completed (DTCE bit is changed from 1 to 0).

In the case of level detection, clear of the IR flag should follow the steps below.

1. Negate the level of an input factor.
2. Run the peripheral read access once and wait for 2 clock cycles of the associated peripheral module clock.
3. Clear the IR flag by writing 0.

Note: During DTCE = 1 writing 0 to IR register is prohibited.

Note: There may be a difference in processing speed between the CPU and ICU, and the CPU may exit the interrupt handler before clearing the IR. Then the CPU will accidentally jump to the interrupt handler again. To avoid this, be sure to read this register before exiting the interrupt handler and make sure that IR is cleared before exiting the interrupt handler.

DTCE bit (DTC Activation Enable)

When the DTCE bit is set to 1, the associated event is selected as the source for DTC activation.

[Setting condition]

- When 1 is written to the DTCE bit.

[Clearing condition]

- When the specified number of transfers is complete. For chain transfers, when the specified number of transfers for the last chain transfer is complete.
- When 0 is written to the DTCE bit.

Note: The secure attribution managed within the Arm CPU NVIC must match the SA (Security Attribution) of IELSRn (n = 0 to 95).

Note: Error during DTC transfer

If an error response occurs during DTC transfer, the DTC notifies the ICU that an error has occurred. ICU clears all bits of the target IELSRn (n = 0 to 95). IELSRn (n = 0 to 95) that is not the target is not cleared.

14.2.18 DELSRm : DMAC Event Link Setting Register m (m = 0 to 7)

Base address: ICU = 0x4000_C000
ICU_NS = 0x5000_C000

Offset address: 0x280 + 0x4 × m

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	IR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	DELS[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	DELS[9:0]	DMAC Event Link Select 0x00: Disable interrupts to the associated DMAC module Others: Event signal number to be linked. For details, see section 14.3.2. Event Number.	R/W*1
15:10	—	These bits are read as 0. The write value should be 0.	R/W
16	IR	DMAC Activation Request Status flag 0: No DMAC activation request occurred 1: DMAC activation request occurred	R/W*2
31:17	—	These bits are read as 0. The write value should be 0.	R/W*1

Note: S-TYPE-3, P-TYPE-2
Note 1. This register must be accessed by halfword or word.
Note 2. Writing 1 to the IR flag is prohibited.

This register has different secure access permission depending on the setting of Trusted Event Route Control Register (TEVTRCR).

If the security attribution is configured as Secure,

- Secure access is allowed.
- Non-secure write access is ignored and Non-secure read access is read as 0, TrustZone access error is generated.

If the security attribution is configured as Non-secure and the trusted event route is disabled,

- Secure write access is ignored and Secure read access is read as 0, TrustZone access error is generated.
- Non-secure access is allowed.

If the security attribution is configured as Non-secure and the trusted event route is enabled,

- Secure access to DELS bit is allowed.
- Non-secure write access to DELS bit is ignored and Non-secure read access to DELS bit is allowed.
- Secure write access to other bits is ignored and Secure read access to other bits is read as 0.
- Non-secure access to other bits is allowed.
- TrustZone access error is not generated.

DELS[9:0] bits (DMAC Event Link Select)

The DELS[9:0] bits link an event signal to the associated DMAC module. All DELS[9:0] bits must be written to simultaneously. Do not set the same event number in multiple DELSRm registers.

[Setting condition]

When value is written to these bits.

[Clearing conditions]

When 0 is written to these bits.

When DMA transfer is stopped by the access error occurs. See section 17, DMA Controller (DMAC) unit chapter.

IR flag (DMAC Activation Request Status flag)

This flag is a status flag for a DMAC activation request.

This flag is associated with the DELS[9:0] bits of this register.

[Setting condition]

When a DMAC activation request occurs from the associated peripheral module or IRQi pin.

[Clearing conditions]

When 0 is written to the flag.

When the DMA transfer is started after a DMAC activation request occurs.

When DMA transfer is stopped by the access error occurs. See section 17, DMA Controller (DMAC).

Note: IR flag is automatically cleared after completion of DMA transfer, so do not write "0" except in case of emergency, such as when an abort occurs. DMA transfer operation when "0" is written during DMA transfer cannot be guaranteed.

14.2.19 WUPEN0 : Wake Up Interrupt Enable Register 0

Base address: ICU = 0x4000_C000
ICU_NS = 0x5000_C000

Offset address: 0x1A0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	RIIC0WUPEN	AGT1CBWUPEN	AGT1CAWUPEN	AGT1UDWUPEN	USBFS0WUPEN	USBHSWUPEN	RTCPRDWUPEN	RTCALMWUPEN	—	—	—	VBATTWUPEN	PVD2WUPEN	PVD1WUPEN	—	IWDTWUPEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	IRQWUPEN15	IRQWUPEN14	IRQWUPEN13	IRQWUPEN12	IRQWUPEN11	IRQWUPEN10	IRQWUPEN9	IRQWUPEN8	IRQWUPEN7	IRQWUPEN6	IRQWUPEN5	IRQWUPEN4	IRQWUPEN3	IRQWUPEN2	IRQWUPEN1	IRQWUPEN0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	IRQWUPEN0 to IRQWUPEN15	Interrupt Software Standby Mode Returns Enable bits from IRQ15 to IRQ0 0: Software Standby Mode returns by IRQn interrupt is disabled. 1: Software Standby Mode returns by IRQn interrupt is enabled*1	R/W
16	IWDTWUPEN	IWDT Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by IWDT interrupt is disabled 1: Software Standby Mode returns by IWDT interrupt is enabled	R/W
17	—	This bit is read as 0. The write value should be 0.	R/W
18	PVD1WUPEN	PVD1 Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by PVD1 interrupt is disabled 1: Software Standby Mode returns by PVD1 interrupt is enabled	R/W
19	PVD2WUPEN	PVD2 Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by PVD2 interrupt is disabled 1: Software Standby Mode returns by PVD2 interrupt is enabled	R/W
20	VBATTWUPEN	VBATT Monitor Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by VBATT monitor interrupt is disabled 1: Software Standby Mode returns by VBATT monitor interrupt is enabled	R/W
23:21	—	These bits are read as 0. The write value should be 0.	R/W
24	RTCALMWUPEN	RTC Alarm Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by RTC alarm interrupt is disabled 1: Software Standby Mode returns by RTC alarm interrupt is enabled	R/W

Bit	Symbol	Function	R/W
25	RTCPRDWUPEN	RTC Period Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by RTC period interrupt is disabled 1: Software Standby Mode returns by RTC period interrupt is enabled	R/W
26	USBHSWUPEN	USBHS Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by USBHS interrupt is disabled 1: Software Standby Mode returns by USBHS interrupt is enabled	R/W
27	USBFS0WUPEN	USBFS Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by USBFS interrupt is disabled 1: Software Standby Mode returns by USBFS interrupt is enabled	R/W
28	AGT1UDWUPEN	AGT1 Underflow Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by AGT1 underflow interrupt is disabled 1: Software Standby Mode returns by AGT1 underflow interrupt is enabled	R/W
29	AGT1CAWUPEN	AGT1 Compare Match A Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by AGT1 compare match A interrupt is disabled 1: Software Standby Mode returns by AGT1 compare match A interrupt is enabled	R/W
30	AGT1CBWUPEN	AGT1 Compare Match B Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by AGT1 compare match B interrupt is disabled 1: Software Standby Mode returns by AGT1 compare match B interrupt is enabled	R/W
31	RIIC0WUPEN	RIIC0 Address Match Interrupt Software Standby Mode Returns Enable bit 0: Software Standby Mode returns by IIC0 address match A interrupt is disabled 1: Software Standby Mode returns by IIC0 address match A interrupt is enabled	R/W

Note: S-TYPE4, P-TYPE2
Note: It is necessary to set to 1 the DSLPWUPIRQENj register bit corresponding to the IELSRn register that selects the factor that enabled the bit of this register.
Note 1. The security attribution of this register is set for each wakeup event. To avoid the occurrence of a security hole, the target event of a wakeup and the security attribution added to this bit must match.

IRQWUPEN15 bit to IRQWUPEN0 bit(Interrupt Software Standby Mode Returns Enable bits from IRQ15 to IRQ0)

The IRQWUPENn (n = 0 to 15) is the enable bit to control the use of the IRQn pin as a Software Standby mode return factor.

IWDTWUPEN bit (IWDT Interrupt Software Standby Mode Returns Enable bit)

The IWDTWUPEN is the enable bit to control the use of the IWDT interrupt as a Software Standby mode return factor.

PVD1WUPEN bit (PVD1 Interrupt Software Standby Mode Returns Enable bit)

The PVD1WUPEN is the enable bit to control the use of the PVD1 interrupt as a Software Standby mode return factor.

PVD2WUPEN bit (PVD2 Interrupt Software Standby Mode Returns Enable bit)

The PVD2WUPEN is the enable bit to control the use of the PVD2 interrupt as a Software Standby mode return factor.

VBATTWUPEN bits (VBATT Monitor Interrupt Software Standby Mode Returns Enable bit)

The VBATTWUPEN is the enable bit to control the use of the VBATT monitor interrupt as a Software Standby mode return factor.

RTCALMWUPEN bits (RTC Alarm Interrupt Software Standby Mode Returns Enable bit)

The RTCALMWUPEN is the enable bit to control the use of the RTC alarm interrupt as a Software Standby mode return factor.

RTCPRDWUPEN bits (RTC Period Interrupt Software Standby Mode Returns Enable bit)

The RTCPRDWUPEN is the enable bit to control the use of the RTC period interrupt as a Software Standby mode return factor.

USBHSWUPEN bits (USBHS Interrupt Software Standby Mode Returns Enable bit)

The USBHSWUPEN is the enable bit to control the use of the USBHS interrupt as a Software Standby mode return factor.

USBFS0WUPEN bits (USBFS Interrupt Software Standby Mode Returns Enable bit)

The USBFS0WUPEN is the enable bit to control the use of the USBFS interrupt as a Software Standby mode return factor.

AGT1UDWUPEN bits (AGT1 Underflow Interrupt Software Standby Mode Returns Enable bit)

The AGT1UDWUPEN is the enable bit to control the use of the AGT1 underflow interrupt as a Software Standby mode return factor.

AGT1CAWUPEN bits (AGT1 Compare Match A Interrupt Software Standby Mode Returns Enable bit)

The AGT1CAWUPEN is the enable bit to control the use of the AGT1 compare match A interrupt as a Software Standby mode return factor.

AGT1CBWUPEN bits (AGT1 Compare Match B Interrupt Software Standby Mode Returns Enable bit)

The AGT1CBWUPEN is the enable bit to control the use of the AGT1 compare match B interrupt as a Software Standby mode return factor.

RIIC0WUPEN bits (RIIC0 Address Match Interrupt Software Standby Mode Returns Enable bit)

The RIIC0WUPEN is the enable bit to control the use of the RIIC0 interrupt as a Software Standby mode return factor.

14.2.20 WUPEN1 : Wake Up Interrupt Enable Register 1

Base address: ICU = 0x4000_C000
ICU_NS = 0x5000_C000

Offset address: 0x1A4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	IRQWUPEN31	IRQWUPEN30	IRQWUPEN29	IRQWUPEN28	IRQWUPEN27	IRQWUPEN26	IRQWUPEN25	IRQWUPEN24	IRQWUPEN23	IRQWUPEN22	IRQWUPEN21	IRQWUPEN20	IRQWUPEN19	IRQWUPEN18	IRQWUPEN17	IRQWUPEN16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PDMWUPEN	ULP1BWUPEN	ULP1AWUPEN	ULP1UWUPEN	I3CWUPEN	ULP0BWUPEN	ULP0AWUPEN	ULP0UWUPEN	SOSCWUPEN	—	—	—	COMPHS0WUPEN	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	—	These bits are read as 0. The write value should be 0.	R/W
3	COMPHS0WUPEN	Comparator-HS0 Interrupt Software Standby Mode returns Enable bit 0: Software Standby returns by Comparator-HS0 interrupt is disabled. 1: Software Standby returns by Comparator-HS0 interrupt is enabled.	R/W
6:4	—	These bits are read as 0. The write value should be 0.	R/W
7	SOSCWUPEN	Sub Oscillation Stop Detection Interrupt Software Standby Returns Enable bit 0: Software Standby returns by Sub Oscillation Stop Detection Interrupt is disabled 1: Software Standby returns by Sub Oscillation Stop Detection Interrupt is enabled	R/W
8	ULP0UWUPEN	ULPT0 Underflow Interrupt Software Standby Mode returns Enable bit 0: Software Standby returns by ULPT0 interrupt is disabled. 1: Software Standby returns by ULPT0 Underflow interrupt is enabled.	R/W
9	ULP0AWUPEN	ULPT0 Compare Match A Interrupt Software Standby Mode returns Enable bit 0: Software Standby returns by ULPT0 Compare match A interrupt is disabled. 1: Software Standby returns by ULPT0 Compare match A interrupt is enabled.	R/W
10	ULP0BWUPEN	ULPT0 Compare Match B Interrupt Software Standby Mode returns Enable bit 0: Software Standby returns by ULPT0 Compare match B interrupt is disabled. 1: Software Standby returns by ULPT0 Compare match B interrupt is enabled.	R/W
11	I3CWUPEN	I3C Wakeup Condition Detection Interrupt Software Standby Mode returns Enable bit 0: Software Standby returns by I3C wake up interrupt is disabled. 1: Software Standby returns by I3C wake up interrupt is enabled.	R/W

Bit	Symbol	Function	R/W
12	ULP1UWUPEN	ULPT1 Underflow Interrupt Software Standby Mode returns Enable bit 0: Software Standby returns by ULPT1 Underflow interrupt is disabled. 1: Software Standby returns by ULPT1 Underflow interrupt is enabled	R/W
13	ULP1AWUPEN	ULPT1 Compare Match A Interrupt Software Standby Mode returns Enable bit 0: Software Standby returns by ULPT1 Compare match A interrupt is disabled. 1: Software Standby returns by ULPT1 Compare match A interrupt is enabled.	R/W
14	ULP1BWUPEN	ULPT1 Compare Match B Interrupt Software Standby Mode returns Enable bit 0: Software Standby returns by ULPT1 Compare match B interrupt is disabled. 1: Software Standby returns by ULPT1 Compare match B interrupt is enabled.	R/W
15	PDMWUPEN	PDMIF Sound Detection Interrupt Software Standby Returns Enable bit 0: Software Standby returns by PDMIF Sound Detection interrupt is disabled 1: Software Standby returns by PDMIF Sound Detection interrupt is enabled	R/W
31:16	IRQWUPEN16 to IRQWUPEN31	Interrupt Software Standby Returns Enable bits IRQ31 to IRQ16 0: Software Standby returns by IRQn interrupt is disabled 1: Software Standby returns by IRQn interrupt is enabled	R/W

Note: S-TYPE4, P-TYPE2

Note: It is necessary to set to 1 the DSLPWUPIRQENj register bit corresponding to the IELSRn register that selects the factor that enabled the bit of this register.

Note: The security attribution of this register is set for each wakeup event. To avoid the occurrence of a security hole, the target event of a wakeup and the security attribution added to this bit must match.

COMPHS0WUPEN bit (Comparator-HS0 Interrupt Software Standby Mode returns Enable bit)

This bit is the enable bit to control the use of the Comparator-HS0 interrupt as a Software Standby return factor.

SOSCWUPEN bit (Sub Oscillation Stop Detection Interrupt Software Standby Returns Enable bit)

This bit is the enable bit to control the use of the Sub Oscillation Stop Detection Interrupt as a Software Standby return factor.

ULP0UWUPEN bit (ULPT0 Underflow Interrupt Software Standby Mode returns Enable bit)

This bit is the enable bit to control the use of the ULPT0 Underflow interrupt as a Software Standby return factor.

ULP0AWUPEN bit (ULPT0 Compare Match A Interrupt Software Standby Mode returns Enable bit)

This bit is the enable bit to control the use of the ULPT0 Compare match A interrupt as a Software Standby return factor.

ULP0BWUPEN bit (ULPT0 Compare Match B Interrupt Software Standby Mode returns Enable bit)

This bit is the enable bit to control the use of the ULPT0 Compare match B interrupt as a Software Standby return factor

I3CWUPEN bit (I3C Wakeup Condition Detection Interrupt Software Standby Mode returns Enable bit)

This bit is the enable bit to control the use of the I3C wake up condition detection interrupt as a Software Standby return factor.

ULP1UWUPEN bit (ULPT1 Underflow Interrupt Software Standby Mode returns Enable bit)

This bit is the enable bit to control the use of the ULPT1 Underflow interrupt as a Software Standby return factor.

ULP1AWUPEN bit (ULPT1 Compare Match A Interrupt Software Standby Mode returns Enable bit)

This bit is the enable bit to control the use of the ULPT1 Compare match A interrupt as a Software Standby return factor.

ULP1BWUPEN bit (ULPT1 Compare Match B Interrupt Software Standby Mode returns Enable bit)

This bit is the enable bit to control the use of the ULPT1 Compare match B interrupt as a Software Standby return factor.

PDMWUPEN bit (PDMIF Sound Detection Interrupt Software Standby Returns Enable bit)

This bit is the enable bit to control the use of the PDMIF Sound Detection interrupt as a Software Standby return factor.

IRQWUPEN31 bit to IRQWUPEN16 bit(Interrupt Software Standby Returns Enable bits IRQ31 to IRQ16)

The IRQWUPEN31 is the enable bit to control the use of the IRQ31 pin as a Software Standby mode return factor.

The IRQWUPEN16 is the enable bit to control the use of the IRQ16 pin as a Software Standby mode return factor.

Bit position and IRQ terminal number correspond

14.2.21 DSLPWUPIRQENj : Deep Sleep Wake Up IRQ Enable Register (j = 0 to 2)

Base address: ICU = 0x4000_C000
ICU_NS = 0x5000_C000

Offset address: 0x210 + 0x4 × j

Bit position:	31 to 0
Bit field:	IRQn (n = 32 × j + 31) to IRQn (n = 32 × j + 0)
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	IRQn	IRQ Deep Sleep Returns Enable bit n. (n = 32 × j + i (i = 0 to 31)) 0: Deep Sleep returns by IRQ event is disabled. 1: Deep Sleep returns by IRQ event is enabled.	R/W

Note: S-TYPE4, P-TYPE2

IRQ bits

This bit is the enable bit to control the use of the IRQ as a Deep Sleep return factor.

The IRQn (n = 32 × j + i (i = 0 to 31)) is selected in the IELSRn.IELS[9:0] bit (n = 0 to 95).

14.2.22 INTSELRp : Interrupt Request Select Register (p = 0 to 31)

Base address: ICU_COMMON = 0x4000_6000
ICU_COMMON_NS = 0x5000_6000

Offset address: 0x40 + 0x4 × p

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	IS32p +31	IS32p +30	IS32p +29	IS32p +28	IS32p +27	IS32p +26	IS32p +25	IS32p +24	IS32p +23	IS32p +22	IS32p +21	IS32p +20	IS32p +19	IS32p +18	IS32p +17	IS32p +16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	IS32p +15	IS32p +14	IS32p +13	IS32p +12	IS32p +11	IS32p +10	IS32p +9	IS32p +8	IS32p +7	IS32p +6	IS32p +5	IS32p +4	IS32p +3	IS32p +2	IS32p +1	IS32p +0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	IS32p+31 to IS32p+0	This bit sets whether the interrupt requests and DTC requests are on the CPU0 side or the CPU1 side. 0: Selected as CPU0 factor 1: Selected as CPU1 factor	R/W*1

Note: S-TYPE-6, P-TYPE-2

Note 1. Bits in the following conditions are fixed to 0 because they are not assigned.

- INTSELR0.IS0 (bit0: event No.0) with no factor assigned.
- Invalid event number with no factor assigned.
For more information, see Column M (Event Func. Assign) and Column Y (CPU Int. Maskable) in the Event List.
- Event number (event No.88,89,91,92,101) unique to CPU0/CPU1.
(DBG CTI1-0 interrupts, IPC CPU mutual interrupt 0, IPC CPU mutual interrupt 1, FPU Exception.)

IS32p + (31 to 0) bits

For example, when IS1 = 0, the cause of Event No.1 is an interrupt for CPU0. Also, when IS2 = 1, the cause of Event No.2 is an interrupt for CPU1.

The relationship between the register and the event number is shown below.

- IS32p+31 to IS32p+0

p = 0 : selected event No.31 to No.0

p = 1 : selected event No.63 to No.32

p = 2 : selected event No.95 to No.64

p = 3 : selected event No.127 to No.96

p = 4 : selected event No.159 to No.128

p = 5 : selected event No.191 to No.160

⋮

p = 14 : selected event No.479 to No.448

p = 15 : selected event No. 511 to No.480

⋮

p = 30: selected event No.991 to No.960

p = 31 : selected event No.1023 to No.992

14.3 Vector Table

The ICU detects maskable and non-maskable interrupts. Interrupt priorities are set up in the Arm NVIC. For information about these registers, see section 14.11. Reference.

14.3.1 Interrupt Vector Table

Table 14.3 describes the interrupt vector table. The interrupt vector addresses conform to the NVIC specifications.

Table 14.3 Interrupt vector table (1 of 4)

Exception number	IRQ number	Vector offset	Source	Description
0	—	0x000	Arm	Initial stack pointer
1	—	0x004	Arm	Initial program counter (reset vector)
2	—	0x008	Arm	Non-Maskable Interrupt (NMI)
3	—	0x00C	Arm	HardFault
4	—	0x010	Arm	MemManage Fault
5	—	0x014	Arm	BusFault
6	—	0x018	Arm	UsageFault
7	—	0x01C	Arm	SecureFault
8	—	0x020	Arm	Reserved
9	—	0x024	Arm	Reserved
10	—	0x028	Arm	Reserved
11	—	0x02C	Arm	Supervisor Call (SVCall)
12	—	0x030	Arm	DebugMonitor
13	—	0x034	Arm	Reserved
14	—	0x038	Arm	Pendable request for system service (PendableSrvReq)
15	—	0x03C	Arm	System Tick Timer (SysTick)
16	0	0x040	ICU.IELSR0	Event selected in the ICU.IELSR0 register
17	1	0x044	ICU.IELSR1	Event selected in the ICU.IELSR1 register
18	2	0x048	ICU.IELSR2	Event selected in the ICU.IELSR2 register
19	3	0x04C	ICU.IELSR3	Event selected in the ICU.IELSR3 register
20	4	0x050	ICU.IELSR4	Event selected in the ICU.IELSR4 register
21	5	0x054	ICU.IELSR5	Event selected in the ICU.IELSR5 register

Table 14.3 Interrupt vector table (2 of 4)

Exception number	IRQ number	Vector offset	Source	Description
22	6	0x058	ICU.IELSR6	Event selected in the ICU.IELSR6 register
23	7	0x05C	ICU.IELSR7	Event selected in the ICU.IELSR7 register
24	8	0x060	ICU.IELSR8	Event selected in the ICU.IELSR8 register
25	9	0x064	ICU.IELSR9	Event selected in the ICU.IELSR9 register
26	10	0x068	ICU.IELSR10	Event selected in the ICU.IELSR10 register
27	11	0x06C	ICU.IELSR11	Event selected in the ICU.IELSR11 register
28	12	0x070	ICU.IELSR12	Event selected in the ICU.IELSR12 register
29	13	0x074	ICU.IELSR13	Event selected in the ICU.IELSR13 register
30	14	0x078	ICU.IELSR14	Event selected in the ICU.IELSR14 register
31	15	0x07C	ICU.IELSR15	Event selected in the ICU.IELSR15 register
32	16	0x080	ICU.IELSR16	Event selected in the ICU.IELSR16 register
33	17	0x084	ICU.IELSR17	Event selected in the ICU.IELSR17 register
34	18	0x088	ICU.IELSR18	Event selected in the ICU.IELSR18 register
35	19	0x08C	ICU.IELSR19	Event selected in the ICU.IELSR19 register
36	20	0x090	ICU.IELSR20	Event selected in the ICU.IELSR20 register
37	21	0x094	ICU.IELSR21	Event selected in the ICU.IELSR21 register
38	22	0x098	ICU.IELSR22	Event selected in the ICU.IELSR22 register
39	23	0x09C	ICU.IELSR23	Event selected in the ICU.IELSR23 register
40	24	0x0A0	ICU.IELSR24	Event selected in the ICU.IELSR24 register
41	25	0x0A4	ICU.IELSR25	Event selected in the ICU.IELSR25 register
42	26	0x0A8	ICU.IELSR26	Event selected in the ICU.IELSR26 register
43	27	0x0AC	ICU.IELSR27	Event selected in the ICU.IELSR27 register
44	28	0x0B0	ICU.IELSR28	Event selected in the ICU.IELSR28 register
45	29	0x0B4	ICU.IELSR29	Event selected in the ICU.IELSR29 register
46	30	0x0B8	ICU.IELSR30	Event selected in the ICU.IELSR30 register
47	31	0x0BC	ICU.IELSR31	Event selected in the ICU.IELSR31 register
48	32	0x0C0	ICU.IELSR32	Event selected in the ICU.IELSR32 register
49	33	0x0C4	ICU.IELSR33	Event selected in the ICU.IELSR33 register
50	34	0x0C8	ICU.IELSR34	Event selected in the ICU.IELSR34 register
51	35	0x0CC	ICU.IELSR35	Event selected in the ICU.IELSR35 register
52	36	0x0D0	ICU.IELSR36	Event selected in the ICU.IELSR36 register
53	37	0x0D4	ICU.IELSR37	Event selected in the ICU.IELSR37 register
54	38	0x0D8	ICU.IELSR38	Event selected in the ICU.IELSR38 register
55	39	0x0DC	ICU.IELSR39	Event selected in the ICU.IELSR39 register
56	40	0x0E0	ICU.IELSR40	Event selected in the ICU.IELSR40 register
57	41	0x0E4	ICU.IELSR41	Event selected in the ICU.IELSR41 register
58	42	0x0E8	ICU.IELSR42	Event selected in the ICU.IELSR42 register
59	43	0x0EC	ICU.IELSR43	Event selected in the ICU.IELSR43 register
60	44	0x0F0	ICU.IELSR44	Event selected in the ICU.IELSR44 register
61	45	0x0F4	ICU.IELSR45	Event selected in the ICU.IELSR45 register
62	46	0x0F8	ICU.IELSR46	Event selected in the ICU.IELSR46 register

Table 14.3 Interrupt vector table (3 of 4)

Exception number	IRQ number	Vector offset	Source	Description
63	47	0x0FC	ICU.IELSR47	Event selected in the ICU.IELSR47 register
64	48	0x100	ICU.IELSR48	Event selected in the ICU.IELSR48 register
65	49	0x104	ICU.IELSR49	Event selected in the ICU.IELSR49 register
66	50	0x108	ICU.IELSR50	Event selected in the ICU.IELSR50 register
67	51	0x10C	ICU.IELSR51	Event selected in the ICU.IELSR51 register
68	52	0x110	ICU.IELSR52	Event selected in the ICU.IELSR52 register
69	53	0x114	ICU.IELSR53	Event selected in the ICU.IELSR53 register
70	54	0x118	ICU.IELSR54	Event selected in the ICU.IELSR54 register
71	55	0x11C	ICU.IELSR55	Event selected in the ICU.IELSR55 register
72	56	0x120	ICU.IELSR56	Event selected in the ICU.IELSR56 register
73	57	0x124	ICU.IELSR57	Event selected in the ICU.IELSR57 register
74	58	0x128	ICU.IELSR58	Event selected in the ICU.IELSR58 register
75	59	0x12C	ICU.IELSR59	Event selected in the ICU.IELSR59 register
76	60	0x130	ICU.IELSR60	Event selected in the ICU.IELSR60 register
77	61	0x134	ICU.IELSR61	Event selected in the ICU.IELSR61 register
78	62	0x138	ICU.IELSR62	Event selected in the ICU.IELSR62 register
79	63	0x13C	ICU.IELSR63	Event selected in the ICU.IELSR63 register
80	64	0x140	ICU.IELSR64	Event selected in the ICU.IELSR64 register
81	65	0x144	ICU.IELSR65	Event selected in the ICU.IELSR65 register
82	66	0x148	ICU.IELSR66	Event selected in the ICU.IELSR66 register
83	67	0x14C	ICU.IELSR67	Event selected in the ICU.IELSR67 register
84	68	0x150	ICU.IELSR68	Event selected in the ICU.IELSR68 register
85	69	0x154	ICU.IELSR69	Event selected in the ICU.IELSR69 register
86	70	0x158	ICU.IELSR70	Event selected in the ICU.IELSR70 register
87	71	0x15C	ICU.IELSR71	Event selected in the ICU.IELSR71 register
88	72	0x160	ICU.IELSR72	Event selected in the ICU.IELSR72 register
89	73	0x164	ICU.IELSR73	Event selected in the ICU.IELSR73 register
90	74	0x168	ICU.IELSR74	Event selected in the ICU.IELSR74 register
91	75	0x16C	ICU.IELSR75	Event selected in the ICU.IELSR75 register
92	76	0x170	ICU.IELSR76	Event selected in the ICU.IELSR76 register
93	77	0x174	ICU.IELSR77	Event selected in the ICU.IELSR77 register
94	78	0x178	ICU.IELSR78	Event selected in the ICU.IELSR78 register
95	79	0x17C	ICU.IELSR79	Event selected in the ICU.IELSR79 register
96	80	0x180	ICU.IELSR80	Event selected in the ICU.IELSR80 register
97	81	0x184	ICU.IELSR81	Event selected in the ICU.IELSR81 register
98	82	0x188	ICU.IELSR82	Event selected in the ICU.IELSR82 register
99	83	0x18C	ICU.IELSR83	Event selected in the ICU.IELSR83 register
100	84	0x190	ICU.IELSR84	Event selected in the ICU.IELSR84 register
101	85	0x194	ICU.IELSR85	Event selected in the ICU.IELSR85 register
102	86	0x198	ICU.IELSR86	Event selected in the ICU.IELSR86 register
103	87	0x19C	ICU.IELSR87	Event selected in the ICU.IELSR87 register

Table 14.3 Interrupt vector table (4 of 4)

Exception number	IRQ number	Vector offset	Source	Description
104	88	0x1A0	ICU.IELSR88	Event selected in the ICU.IELSR88 register
105	89	0x1A4	ICU.IELSR89	Event selected in the ICU.IELSR89 register
106	90	0x1A8	ICU.IELSR90	Event selected in the ICU.IELSR90 register
107	91	0x1AC	ICU.IELSR91	Event selected in the ICU.IELSR91 register
108	92	0x1B0	ICU.IELSR92	Event selected in the ICU.IELSR92 register
109	93	0x1B4	ICU.IELSR93	Event selected in the ICU.IELSR93 register
110	94	0x1B8	ICU.IELSR94	Event selected in the ICU.IELSR94 register
111	95	0x1BC	ICU.IELSR95	Event selected in the ICU.IELSR95 register

14.3.2 Event Number

The following table lists heading details for Table 14.5, which describes each event number.

Table 14.4 Heading details for Event List

Heading	Description
Interrupt request source	Name of the source generating the interrupt request
Name	Name of the interrupt
Connect to NVIC	" ✓" indicates the interrupt can be used as a CPU interrupt
Invoke DTC	" ✓" indicates the interrupt can be used to request DTC activation
Invoke DMAC	"✓" indicates the interrupt can be used to request DMAC activation
Canceling CPU Deep Sleep	"✓" indicates the interrupt can be used to request a return from CPU Deep Sleep mode
Canceling Software Standby	" ✓" indicates the interrupt can be used to request a return from Software Standby mode
Canceling Deep Software Standby	"✓" indicates the interrupt can be used to request a return from Deep Software Standby mode

Table 14.5 Event List (1 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x001	Port	PORT_IRQ0	✓	✓	✓	✓	✓	✓
0x002		PORT_IRQ1	✓	✓	✓	✓	✓	✓
0x003		PORT_IRQ2	✓	✓	✓	✓	✓	✓
0x004		PORT_IRQ3	✓	✓	✓	✓	✓	✓
0x005		PORT_IRQ4	✓	✓	✓	✓	✓	✓
0x006		PORT_IRQ5	✓	✓	✓	✓	✓	✓
0x007		PORT_IRQ6	✓	✓	✓	✓	✓	✓
0x008		PORT_IRQ7	✓	✓	✓	✓	✓	✓
0x009		PORT_IRQ8	✓	✓	✓	✓	✓	✓
0x00A		PORT_IRQ9	✓	✓	✓	✓	✓	✓
0x00B		PORT_IRQ10	✓	✓	✓	✓	✓	✓
0x00C		PORT_IRQ11	✓	✓	✓	✓	✓	✓
0x00D		PORT_IRQ12	✓	✓	✓	✓	✓	✓
0x00E		PORT_IRQ13	✓	✓	✓	✓	✓	✓
0x00F		PORT_IRQ14	✓	✓	✓	✓	✓	✓
0x010		PORT_IRQ15	✓	✓	✓	✓	✓	✓
0x011		PORT_IRQ16	✓	✓	✓	✓	✓	✓
0x012		PORT_IRQ17	✓	✓	✓	✓	✓	✓
0x013		PORT_IRQ18	✓	✓	✓	✓	✓	✓
0x014		PORT_IRQ19	✓	✓	✓	✓	✓	✓
0x015		PORT_IRQ20	✓	✓	✓	✓	✓	✓
0x016		PORT_IRQ21	✓	✓	✓	✓	✓	✓
0x017		PORT_IRQ22	✓	✓	✓	✓	✓	✓
0x018		PORT_IRQ23	✓	✓	✓	✓	✓	✓
0x019		PORT_IRQ24	✓	✓	✓	✓	✓	✓
0x01A		PORT_IRQ25	✓	✓	✓	✓	✓	✓
0x01B		PORT_IRQ26	✓	✓	✓	✓	✓	✓
0x01C		PORT_IRQ27	✓	✓	✓	✓	✓	✓
0x01D		PORT_IRQ28	✓	✓	✓	✓	✓	✓
0x01E		PORT_IRQ29	✓	✓	✓	✓	✓	✓
0x01F		PORT_IRQ30	✓	✓	✓	✓	✓	✓
0x020		PORT_IRQ31	✓	✓	✓	✓	✓	✓
0x040	DMAC00	DMAC00_INT	✓	✓	—	✓	—	—
0x041	DMAC01	DMAC01_INT	✓	✓	—	✓	—	—
0x042	DMAC02	DMAC02_INT	✓	✓	—	✓	—	—
0x043	DMAC03	DMAC03_INT	✓	✓	—	✓	—	—
0x044	DMAC04	DMAC04_INT	✓	✓	—	✓	—	—
0x045	DMAC05	DMAC05_INT	✓	✓	—	✓	—	—
0x046	DMAC06	DMAC06_INT	✓	✓	—	✓	—	—
0x047	DMAC07	DMAC07_INT	✓	✓	—	✓	—	—

Table 14.5 Event List (2 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x048	DMAC10	DMAC10_INT	✓	✓	—	✓	—	—
0x049	DMAC11	DMAC11_INT	✓	✓	—	✓	—	—
0x04A	DMAC12	DMAC12_INT	✓	✓	—	✓	—	—
0x04B	DMAC13	DMAC13_INT	✓	✓	—	✓	—	—
0x04C	DMAC14	DMAC14_INT	✓	✓	—	✓	—	—
0x04D	DMAC15	DMAC15_INT	✓	✓	—	✓	—	—
0x04E	DMAC16	DMAC16_INT	✓	✓	—	✓	—	—
0x04F	DMAC17	DMAC17_INT	✓	✓	—	✓	—	—
0x056	DMAC/DTC	DMA0_TRANSERR	✓	—	—	✓	—	—
0x057	DMAC/DTC	DMA1_TRANSERR	✓	—	—	✓	—	—
0x058	DBG	DBG_CTIIRQ0	✓	—	—	—	—	—
0x059		DBG_CTIIRQ1	✓	—	—	—	—	—
0x05A		DBG_JBRXI	✓	—	—	✓	—	—
0x05B	IPC	IPC_IRQ0	✓	—	—	✓	—	—
0x05C		IPC_IRQ1	✓	—	—	✓	—	—
0x05F	Local Memory	LM0_ERR	✓	—	—	✓	—	—
0x060		LM1_ERR	✓	—	—	✓	—	—
0x061	CPU	CPU0_LOCKUP	✓	—	—	✓	—	—
0x062		CPU1_LOCKUP	✓	—	—	✓	—	—
0x063	BUS	BUS_ERR	✓	—	—	✓	—	—
0x064	CM	CM_ERR	✓	—	—	✓	—	—
0x065	FPU	FPU_EXC	✓	—	—	—	—	—
0x067	NPU	NPU_IRQ	✓	—	—	✓	—	—
0x068	MRAM	MRAM_MRCRD	✓	—	—	✓	—	—
0x069		MRAM_MRERD	✓	—	—	✓	—	—
0x06B		MRAM_MRCPR	✓	—	—	✓	—	—
0x06C		MRAM_MREPR	✓	—	—	✓	—	—
0x06D		MRAM_ENDOFPE	✓	—	—	✓	—	—
0x070	PVD1	PVD_PVD1	✓	—	—	✓	✓	✓*4
0x071	PVD2	PVD_PVD2	✓	—	—	✓	✓	✓*4
0x075	BBF	VBATT_TADI	✓	—	—	✓	✓	✓
0x076	MOSC	MOSC_STOP	✓	—	—	✓	—	—
0x077	SOSC	SOSC_STOP	✓	—	—	✓	✓	✓
0x080	ULPT0	ULPT0_ULPTI	✓	✓	✓	✓	✓	✓*5
0x081		ULPT0_ULPTCMAI	✓	✓	✓	✓	✓	—
0x082		ULPT0_ULPTCMBI	✓	✓	✓	✓	✓	—

Table 14.5 Event List (3 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x083	ULPT1	ULPT1_ULPTI	✓	✓	✓	✓	✓	✓ *5
0x084		ULPT1_ULPTCMAI	✓	✓	✓	✓	✓	—
0x085		ULPT1_ULPTCMBI	✓	✓	✓	✓	✓	—
0x086	AGT0	AGT0_AGTI	✓	✓	✓	✓	—	—
0x087		AGT0_AGTCMAI	✓	✓	✓	✓	—	—
0x088		AGT0_AGTCMBI	✓	✓	✓	✓	—	—
0x089	AGT1	AGT1_AGTI	✓	✓	✓	✓	✓	—
0x08A		AGT1_AGTCMAI	✓	✓	✓	✓	✓	—
0x08B		AGT1_AGTCMBI	✓	✓	✓	✓	✓	—
0x092	IWDT	IWDT_NMIUNDF	✓	—	—	✓	✓	✓ *5
0x093	WDT0	WDT0_NMIUNDF	✓	—	—	✓	—	—
0x094	WDT1	WDT1_NMIUNDF	✓	—	—	✓	—	—
0x095	RTC	RTC_ALM	✓	—	—	✓	✓	✓
0x096		RTC_PRD	✓	—	—	✓	✓	✓
0x097		RTC_CUP	✓	—	—	✓	—	—
0x098	USBFS	USBFS_D0FIFO	✓	✓	✓	✓	—	—
0x099		USBFS_D1FIFO	✓	✓	✓	✓	—	—
0x09A		USBFS_USBI	✓	—	—	✓	—	—
0x09B		USBFS_USBR	✓	—	—	✓	✓	✓ *5
0x09C	IIC0	IIC0_RXI	✓	✓	✓	✓	—	—
0x09D		IIC0_TXI	✓	✓	✓	✓	—	—
0x09E		IIC0_TEI	✓	—	—	✓	—	—
0x09F		IIC0_EEI	✓	—	—	✓	—	—
0x0A0		IIC0_WUI	✓	—	—	✓	✓	—
0x0A1	IIC1	IIC1_RXI	✓	✓	✓	✓	—	—
0x0A2		IIC1_TXI	✓	✓	✓	✓	—	—
0x0A3		IIC1_TEI	✓	—	—	✓	—	—
0x0A4		IIC1_EEI	✓	—	—	✓	—	—
0x0A6	IIC2	IIC2_RXI	✓	✓	✓	✓	—	—
0x0A7		IIC2_TXI	✓	✓	✓	✓	—	—
0x0A8		IIC2_TEI	✓	—	—	✓	—	—
0x0A9		IIC2_EEI	✓	—	—	✓	—	—
0x0AB	SDHI/ MMC0	SDHI_MMC0_ACCS	✓	—	—	✓	—	—
0x0AC		SDHI_MMC0_SDIO	✓	—	—	✓	—	—
0x0AD		SDHI_MMC0_CARD	✓	—	—	✓	—	—
0x0AE		SDHI_MMC0_ODMSDBREQ	—	✓	✓	✓	—	—
0x0AF	SDHI/ MMC1	SDHI_MMC1_ACCS	✓	—	—	✓	—	—
0x0B0		SDHI_MMC1_SDIO	✓	—	—	✓	—	—
0x0B1		SDHI_MMC1_CARD	✓	—	—	✓	—	—
0x0B2		SDHI_MMC1_ODMSDBREQ	—	✓	✓	✓	—	—

Table 14.5 Event List (4 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x0B3	SSI0	SSI0_SSITXI	✓	✓	✓	✓	—	—
0x0B4		SSI0_SSIRXI	✓	✓	✓	✓	—	—
0x0B6		SSI0_SSIF	✓	—	—	✓	—	—
0x0B9	SSI1	SSI1_SSIRT	✓	✓	✓	✓	—	—
0x0BA		SSI1_SSIF	✓	—	—	✓	—	—
0x0BB	OSPI0	OSPI0_ERR	✓	—	—	✓	—	—
0x0BC		OSPI0_CMP	✓	—	—	✓	—	—
0x0BD	OSPI1	OSPI1_ERR	✓	—	—	✓	—	—
0x0BE		OSPI1_CMP	✓	—	—	✓	—	—
0x0BF	PDMIF	PDM_DAT0	✓	✓	✓	✓	—	—
0x0C0		PDM_DAT1	✓	✓	✓	✓	—	—
0x0C1		PDM_DAT2	✓	✓	✓	✓	—	—
0x0C2		PDM_SDET	✓	—	—	✓	✓	—
0x0C3		PDM_ERR0	✓	—	—	✓	—	—
0x0C4		PDM_ERR1	✓	—	—	✓	—	—
0x0C5		PDM_ERR2	✓	—	—	✓	—	—
0x0C6	ACMPHS	ACMP_HS0	✓	—	—	✓*1	✓*1	—
0x0C7		ACMP_HS1	✓	—	—	✓	—	—
0x0C8		ACMP_HS2	✓	—	—	✓	—	—
0x0C9		ACMP_HS3	✓	—	—	✓	—	—
0x0CC	ELC	ELC_SWEVT0	✓*3	✓	—	✓	—	—
0x0CD		ELC_SWEVT1	✓*3	✓	—	✓	—	—
0x0CE		ELC_SWEVT2	✓*3	✓	—	✓	—	—
0x0CF		ELC_SWEVT3	✓*3	✓	—	✓	—	—
0x0D0	PORT	IOPORT_GROUP1	✓	✓*2	✓*2	✓	—	—
0x0D1		IOPORT_GROUP2	✓	✓*2	✓*2	✓	—	—
0x0D2		IOPORT_GROUP3	✓	✓*2	✓*2	✓	—	—
0x0D3		IOPORT_GROUP4	✓	✓*2	✓*2	✓	—	—
0x0D4	CAC	CAC_FEERI	✓	—	—	✓	—	—
0x0D5		CAC_MENDI	✓	—	—	✓	—	—
0x0D6		CAC_OVFI	✓	—	—	✓	—	—
0x0D7	POEG	POEG_GROUPA	✓	—	—	✓	—	—
0x0D8		POEG_GROUPB	✓	—	—	✓	—	—
0x0D9		POEG_GROUPC	✓	—	—	✓	—	—
0x0DA		POEG_GROUPD	✓	—	—	✓	—	—
0x180	GPT	GPT_UVWEDGE	✓	—	—	✓	—	—

Table 14.5 Event List (5 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x181	GPT0	GPT0_CCMPA	✓	✓	✓	✓	—	—
0x182		GPT0_CCMPB	✓	✓	✓	✓	—	—
0x183		GPT0_CMPC	✓	✓	✓	✓	—	—
0x184		GPT0_CMPD	✓	✓	✓	✓	—	—
0x185		GPT0_CMPE	✓	✓	✓	✓	—	—
0x186		GPT0_CMPF	✓	✓	✓	✓	—	—
0x187		GPT0_OVF	✓	✓	✓	✓	—	—
0x188		GPT0_UDF	✓	✓	✓	✓	—	—
0x189		GPT0_PC	✓	✓	✓	✓	—	—
0x18A	GPT1	GPT1_CCMPA	✓	✓	✓	✓	—	—
0x18B		GPT1_CCMPB	✓	✓	✓	✓	—	—
0x18C		GPT1_CMPC	✓	✓	✓	✓	—	—
0x18D		GPT1_CMPD	✓	✓	✓	✓	—	—
0x18E		GPT1_CMPE	✓	✓	✓	✓	—	—
0x18F		GPT1_CMPF	✓	✓	✓	✓	—	—
0x190		GPT1_OVF	✓	✓	✓	✓	—	—
0x191		GPT1_UDF	✓	✓	✓	✓	—	—
0x192		GPT1_PC	✓	✓	✓	✓	—	—
0x193	GPT2	GPT2_CCMPA	✓	✓	✓	✓	—	—
0x194		GPT2_CCMPB	✓	✓	✓	✓	—	—
0x195		GPT2_CMPC	✓	✓	✓	✓	—	—
0x196		GPT2_CMPD	✓	✓	✓	✓	—	—
0x197		GPT2_CMPE	✓	✓	✓	✓	—	—
0x198		GPT2_CMPF	✓	✓	✓	✓	—	—
0x199		GPT2_OVF	✓	✓	✓	✓	—	—
0x19A		GPT2_UDF	✓	✓	✓	✓	—	—
0x19B		GPT2_PC	✓	✓	✓	✓	—	—
0x19C	GPT3	GPT3_CCMPA	✓	✓	✓	✓	—	—
0x19D		GPT3_CCMPB	✓	✓	✓	✓	—	—
0x19E		GPT3_CMPC	✓	✓	✓	✓	—	—
0x19F		GPT3_CMPD	✓	✓	✓	✓	—	—
0x1A0		GPT3_CMPE	✓	✓	✓	✓	—	—
0x1A1		GPT3_CMPF	✓	✓	✓	✓	—	—
0x1A2		GPT3_OVF	✓	✓	✓	✓	—	—
0x1A3		GPT3_UDF	✓	✓	✓	✓	—	—
0x1A4		GPT3_PC	✓	✓	✓	✓	—	—

Table 14.5 Event List (6 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x1A5	GPT4	GPT4_CCMPA	✓	✓	✓	✓	—	—
0x1A6		GPT4_CCMPB	✓	✓	✓	✓	—	—
0x1A7		GPT4_CMPC	✓	✓	✓	✓	—	—
0x1A8		GPT4_CMPD	✓	✓	✓	✓	—	—
0x1A9		GPT4_CMPE	✓	✓	✓	✓	—	—
0x1AA		GPT4_CMPF	✓	✓	✓	✓	—	—
0x1AB		GPT4_OVF	✓	✓	✓	✓	—	—
0x1AC		GPT4_UDF	✓	✓	✓	✓	—	—
0x1AE	GPT5	GPT5_CCMPA	✓	✓	✓	✓	—	—
0x1AF		GPT5_CCMPB	✓	✓	✓	✓	—	—
0x1B0		GPT5_CMPC	✓	✓	✓	✓	—	—
0x1B1		GPT5_CMPD	✓	✓	✓	✓	—	—
0x1B2		GPT5_CMPE	✓	✓	✓	✓	—	—
0x1B3		GPT5_CMPF	✓	✓	✓	✓	—	—
0x1B4		GPT5_OVF	✓	✓	✓	✓	—	—
0x1B5		GPT5_UDF	✓	✓	✓	✓	—	—
0x1B7	GPT6	GPT6_CCMPA	✓	✓	✓	✓	—	—
0x1B8		GPT6_CCMPB	✓	✓	✓	✓	—	—
0x1B9		GPT6_CMPC	✓	✓	✓	✓	—	—
0x1BA		GPT6_CMPD	✓	✓	✓	✓	—	—
0x1BB		GPT6_CMPE	✓	✓	✓	✓	—	—
0x1BC		GPT6_CMPF	✓	✓	✓	✓	—	—
0x1BD		GPT6_OVF	✓	✓	✓	✓	—	—
0x1BE		GPT6_UDF	✓	✓	✓	✓	—	—
0x1C0	GPT7	GPT7_CCMPA	✓	✓	✓	✓	—	—
0x1C1		GPT7_CCMPB	✓	✓	✓	✓	—	—
0x1C2		GPT7_CMPC	✓	✓	✓	✓	—	—
0x1C3		GPT7_CMPD	✓	✓	✓	✓	—	—
0x1C4		GPT7_CMPE	✓	✓	✓	✓	—	—
0x1C5		GPT7_CMPF	✓	✓	✓	✓	—	—
0x1C6		GPT7_OVF	✓	✓	✓	✓	—	—
0x1C7		GPT7_UDF	✓	✓	✓	✓	—	—
0x1C9	GPT8	GPT8_CCMPA	✓	✓	✓	✓	—	—
0x1CA		GPT8_CCMPB	✓	✓	✓	✓	—	—
0x1CB		GPT8_CMPC	✓	✓	✓	✓	—	—
0x1CC		GPT8_CMPD	✓	✓	✓	✓	—	—
0x1CD		GPT8_CMPE	✓	✓	✓	✓	—	—
0x1CE		GPT8_CMPF	✓	✓	✓	✓	—	—
0x1CF		GPT8_OVF	✓	✓	✓	✓	—	—
0x1D0		GPT8_UDF	✓	✓	✓	✓	—	—

Table 14.5 Event List (7 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x1D2	GPT9	GPT9_CCMPA	✓	✓	✓	✓	—	—
0x1D3		GPT9_CCMPB	✓	✓	✓	✓	—	—
0x1D4		GPT9_CMPC	✓	✓	✓	✓	—	—
0x1D5		GPT9_CMPD	✓	✓	✓	✓	—	—
0x1D6		GPT9_CMPE	✓	✓	✓	✓	—	—
0x1D7		GPT9_CMPF	✓	✓	✓	✓	—	—
0x1D8		GPT9_OVF	✓	✓	✓	✓	—	—
0x1D9		GPT9_UDF	✓	✓	✓	✓	—	—
0x1DB	GPT10	GPT10_CCMPA	✓	✓	✓	✓	—	—
0x1DC		GPT10_CCMPB	✓	✓	✓	✓	—	—
0x1DD		GPT10_CMPC	✓	✓	✓	✓	—	—
0x1DE		GPT10_CMPD	✓	✓	✓	✓	—	—
0x1DF		GPT10_CMPE	✓	✓	✓	✓	—	—
0x1E0		GPT10_CMPF	✓	✓	✓	✓	—	—
0x1E1		GPT10_OVF	✓	✓	✓	✓	—	—
0x1E2		GPT10_UDF	✓	✓	✓	✓	—	—
0x1E3		GPT10_PC	✓	✓	✓	✓	—	—
0x1E4	GPT11	GPT11_CCMPA	✓	✓	✓	✓	—	—
0x1E5		GPT11_CCMPB	✓	✓	✓	✓	—	—
0x1E6		GPT11_CMPC	✓	✓	✓	✓	—	—
0x1E7		GPT11_CMPD	✓	✓	✓	✓	—	—
0x1E8		GPT11_CMPE	✓	✓	✓	✓	—	—
0x1E9		GPT11_CMPF	✓	✓	✓	✓	—	—
0x1EA		GPT11_OVF	✓	✓	✓	✓	—	—
0x1EB		GPT11_UDF	✓	✓	✓	✓	—	—
0x1EC		GPT11_PC	✓	✓	✓	✓	—	—
0x1ED	GPT12	GPT12_CCMPA	✓	✓	✓	✓	—	—
0x1EE		GPT12_CCMPB	✓	✓	✓	✓	—	—
0x1EF		GPT12_CMPC	✓	✓	✓	✓	—	—
0x1F0		GPT12_CMPD	✓	✓	✓	✓	—	—
0x1F1		GPT12_CMPE	✓	✓	✓	✓	—	—
0x1F2		GPT12_CMPF	✓	✓	✓	✓	—	—
0x1F3		GPT12_OVF	✓	✓	✓	✓	—	—
0x1F4		GPT12_UDF	✓	✓	✓	✓	—	—
0x1F5		GPT12_PC	✓	✓	✓	✓	—	—

Table 14.5 Event List (8 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x1F6	GPT13	GPT13_CCMPA	✓	✓	✓	✓	—	—
0x1F7		GPT13_CCMPB	✓	✓	✓	✓	—	—
0x1F8		GPT13_CMPC	✓	✓	✓	✓	—	—
0x1F9		GPT13_CMPD	✓	✓	✓	✓	—	—
0x1FA		GPT13_CMPE	✓	✓	✓	✓	—	—
0x1FB		GPT13_CMPF	✓	✓	✓	✓	—	—
0x1FC		GPT13_OVF	✓	✓	✓	✓	—	—
0x1FD		GPT13_UDF	✓	✓	✓	✓	—	—
0x1FE		GPT13_PC	✓	✓	✓	✓	—	—
0x29A	ESWM	ETHER_FWEI	✓	—	—	✓	—	—
0x29B		ETHER_CAEI	✓	—	—	✓	—	—
0x29C		ETHER_GWEI	✓	—	—	✓	—	—
0x29D		ETHER_EAEI0	✓	—	—	✓	—	—
0x29E		ETHER_EAEI1	✓	—	—	✓	—	—
0x29F		ETHER_PTPSI0	✓	—	—	✓	—	—
0x2A0		ETHER_PTPSI1	✓	—	—	✓	—	—
0x2A1		ETHER_FWSI	✓	—	—	✓	—	—
0x2A3		ETHER_CAMI	✓	—	—	✓	—	—
0x2A4		ETHER_EASI0	✓	—	—	✓	—	—
0x2A5		ETHER_EASI1	✓	—	—	✓	—	—
0x2A6		ETHER_GWDI0	✓	—	—	✓	—	—
0x2A7		ETHER_GWDI1	✓	—	—	✓	—	—
0x2A8		ETHER_GWDI2	✓	—	—	✓	—	—
0x2A9		ETHER_GWDI3	✓	—	—	✓	—	—
0x2AA		ETHER_GWDI4	✓	—	—	✓	—	—
0x2AB		ETHER_GWDI5	✓	—	—	✓	—	—
0x2AC		ETHER_GWDI6	✓	—	—	✓	—	—
0x2AD		ETHER_GWDI7	✓	—	—	✓	—	—
0x2AE		ETHER_TSDI0	✓	—	—	✓	—	—
0x2AF		ETHER_TSDI1	✓	—	—	✓	—	—
0x2B0		ETHER_MDIO0	✓	—	—	✓	—	—
0x2B1		ETHER_MDIO1	✓	—	—	✓	—	—
0x2B2		ETHER_RMPI0	✓	—	—	✓	—	—
0x2B3		ETHER_RMPI1	✓	—	—	✓	—	—
0x2B4		GPTP_PTPOUT0	✓	✓	✓	✓	—	—
0x2B5		GPTP_PTPOUT1	✓	✓	✓	✓	—	—
0x2B6		GPTP_PTPOUT2	✓	✓	✓	✓	—	—
0x2B7		GPTP_PTPOUT3	✓	✓	✓	✓	—	—
0x2B8		GPTP_MATCH0	✓	✓	✓	✓	—	—
0x2B9		GPTP_MATCH1	✓	✓	✓	✓	—	—

Table 14.5 Event List (9 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x2C1	USBHS	USBHS_D0FIFO	✓	✓	✓	✓	—	—
0x2C2		USBHS_D1FIFO	✓	✓	✓	✓	—	—
0x2C3		USBHS_USBIR	✓	—	—	✓	✓	✓*5
0x2C4	SCI0	SCI0_RXI	✓	✓	✓	✓	—	—
0x2C5		SCI0_TXI	✓	✓	✓	✓	—	—
0x2C6		SCI0_TEI	✓	—	—	✓	—	—
0x2C7		SCI0_ERI	✓	—	—	✓	—	—
0x2C8		SCI0_AED	✓	—	—	✓	—	—
0x2C9		SCI0_BFD	✓	—	—	✓	—	—
0x2CA		SCI0_AM	✓	—	—	✓	—	—
0x2CB	SCI1	SCI1_RXI	✓	✓	✓	✓	—	—
0x2CC		SCI1_TXI	✓	✓	✓	✓	—	—
0x2CD		SCI1_TEI	✓	—	—	✓	—	—
0x2CE		SCI1_ERI	✓	—	—	✓	—	—
0x2CF		SCI1_AED	✓	—	—	✓	—	—
0x2D0		SCI1_BFD	✓	—	—	✓	—	—
0x2D1		SCI1_AM	✓	—	—	✓	—	—
0x2D2	SCI2	SCI2_RXI	✓	✓	✓	✓	—	—
0x2D3		SCI2_TXI	✓	✓	✓	✓	—	—
0x2D4		SCI2_TEI	✓	—	—	✓	—	—
0x2D5		SCI2_ERI	✓	—	—	✓	—	—
0x2D6		SCI2_AED	✓	—	—	✓	—	—
0x2D7		SCI2_BFD	✓	—	—	✓	—	—
0x2D8		SCI2_AM	✓	—	—	✓	—	—
0x2D9	SCI3	SCI3_RXI	✓	✓	✓	✓	—	—
0x2DA		SCI3_TXI	✓	✓	✓	✓	—	—
0x2DB		SCI3_TEI	✓	—	—	✓	—	—
0x2DC		SCI3_ERI	✓	—	—	✓	—	—
0x2DD		SCI3_AED	✓	—	—	✓	—	—
0x2DE		SCI3_BFD	✓	—	—	✓	—	—
0x2DF		SCI3_AM	✓	—	—	✓	—	—
0x2E0	SCI4	SCI4_RXI	✓	✓	✓	✓	—	—
0x2E1		SCI4_TXI	✓	✓	✓	✓	—	—
0x2E2		SCI4_TEI	✓	—	—	✓	—	—
0x2E3		SCI4_ERI	✓	—	—	✓	—	—
0x2E4		SCI4_AED	✓	—	—	✓	—	—
0x2E5		SCI4_BFD	✓	—	—	✓	—	—
0x2E6		SCI4_AM	✓	—	—	✓	—	—

Table 14.5 Event List (10 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x2E7	SCI5	SCI5_RXI	✓	✓	✓	✓	—	—
0x2E8		SCI5_TXI	✓	✓	✓	✓	—	—
0x2E9		SCI5_TEI	✓	—	—	✓	—	—
0x2EA		SCI5_ERI	✓	—	—	✓	—	—
0x2EB		SCI5_AED	✓	—	—	✓	—	—
0x2EC		SCI5_BFD	✓	—	—	✓	—	—
0x2ED		SCI5_AM	✓	—	—	✓	—	—
0x2EE	SCI6	SCI6_RXI	✓	✓	✓	✓	—	—
0x2EF		SCI6_TXI	✓	✓	✓	✓	—	—
0x2F0		SCI6_TEI	✓	—	—	✓	—	—
0x2F1		SCI6_ERI	✓	—	—	✓	—	—
0x2F2		SCI6_AED	✓	—	—	✓	—	—
0x2F3		SCI6_BFD	✓	—	—	✓	—	—
0x2F4		SCI6_AM	✓	—	—	✓	—	—
0x2F5	SCI7	SCI7_RXI	✓	✓	✓	✓	—	—
0x2F6		SCI7_TXI	✓	✓	✓	✓	—	—
0x2F7		SCI7_TEI	✓	—	—	✓	—	—
0x2F8		SCI7_ERI	✓	—	—	✓	—	—
0x2F9		SCI7_AED	✓	—	—	✓	—	—
0x2FA		SCI7_BFD	✓	—	—	✓	—	—
0x2FB		SCI7_AM	✓	—	—	✓	—	—
0x2FC	SCI8	SCI8_RXI	✓	✓	✓	✓	—	—
0x2FD		SCI8_TXI	✓	✓	✓	✓	—	—
0x2FE		SCI8_TEI	✓	—	—	✓	—	—
0x2FF		SCI8_ERI	✓	—	—	✓	—	—
0x300		SCI8_AED	✓	—	—	✓	—	—
0x301		SCI8_BFD	✓	—	—	✓	—	—
0x302		SCI8_AM	✓	—	—	✓	—	—
0x303	SCI9	SCI9_RXI	✓	✓	✓	✓	—	—
0x304		SCI9_TXI	✓	✓	✓	✓	—	—
0x305		SCI9_TEI	✓	—	—	✓	—	—
0x306		SCI9_ERI	✓	—	—	✓	—	—
0x307		SCI9_AED	✓	—	—	✓	—	—
0x308		SCI9_BFD	✓	—	—	✓	—	—
0x309		SCI9_AM	✓	—	—	✓	—	—

Table 14.5 Event List (11 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x30A	SCI10	SCI10_RXI	✓	✓	✓	✓	—	—
0x30B		SCI10_TXI	✓	✓	✓	✓	—	—
0x30C		SCI10_TEI	✓	—	—	✓	—	—
0x30D		SCI10_ERI	✓	—	—	✓	—	—
0x30E		SCI10_AED	✓	—	—	✓	—	—
0x30F		SCI10_BFD	✓	—	—	✓	—	—
0x310		SCI10_AM	✓	—	—	✓	—	—
0x311	SCI11	SCI11_RXI	✓	✓	✓	✓	—	—
0x312		SCI11_TXI	✓	✓	✓	✓	—	—
0x313		SCI11_TEI	✓	—	—	✓	—	—
0x314		SCI11_ERI	✓	—	—	✓	—	—
0x315		SCI11_AED	✓	—	—	✓	—	—
0x316		SCI11_BFD	✓	—	—	✓	—	—
0x317		SCI11_AM	✓	—	—	✓	—	—
0x318	SPI0	SPI0_SPRI	✓	✓	✓	✓	—	—
0x319		SPI0_SPTI	✓	✓	✓	✓	—	—
0x31A		SPI0_SPII	✓	—	—	✓	—	—
0x31B		SPI0_SPEI	✓	—	—	✓	—	—
0x31C		SPI0_SPCEND	✓	—	—	✓	—	—
0x31D	SPI1	SPI1_SPRI	✓	✓	✓	✓	—	—
0x31E		SPI1_SPTI	✓	✓	✓	✓	—	—
0x31F		SPI1_SPII	✓	—	—	✓	—	—
0x320		SPI1_SPEI	✓	—	—	✓	—	—
0x321		SPI1_SPCEND	✓	—	—	✓	—	—

Table 14.5 Event List (12 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x322	CANFD	CAN_RXF	✓	—	—	✓	—	—
0x323		CAN_GLERR	✓	—	—	✓	—	—
0x324		CAN0_RF_DMAREQ0	✓	✓	✓	✓	—	—
0x325		CAN0_RF_DMAREQ1	✓	✓	✓	✓	—	—
0x328		CAN1_RF_DMAREQ0	✓	✓	✓	✓	—	—
0x329		CAN1_RF_DMAREQ1	✓	✓	✓	✓	—	—
0x32C		CAN0_TX	✓	—	—	✓	—	—
0x32D		CAN0_CHERR	✓	—	—	✓	—	—
0x32E		CAN0_COMFRX	✓	—	—	✓	—	—
0x32F		CAN0_CF_DMAREQ	✓	✓	✓	✓	—	—
0x330		CAN0_RXMB	✓	—	—	✓	—	—
0x331		CAN1_TX	✓	—	—	✓	—	—
0x332		CAN1_CHERR	✓	—	—	✓	—	—
0x333		CAN1_COMFRX	✓	—	—	✓	—	—
0x334		CAN1_CF_DMAREQ	✓	✓	✓	✓	—	—
0x335		CAN1_RXMB	✓	—	—	✓	—	—
0x338		CAN0_MRAM_ERI	✓	—	—	✓	—	—
0x339		CAN1_MRAM_ERI	✓	—	—	✓	—	—
0x33A	I3C	I3C_RESP	✓	✓	✓	✓	—	—
0x33B		I3C_CMD	✓	✓	✓	✓	—	—
0x33C		I3C_IBI	✓	✓	✓	✓	—	—
0x33D		I3C_RX	✓	✓	✓	✓	—	—
0x33E		I3C_TX	✓	✓	✓	✓	—	—
0x33F		I3C_RCV	✓	✓	✓	✓	—	—
0x340		I3C_HRESP	✓	✓	✓	✓	—	—
0x341		I3C_HCMD	✓	✓	✓	✓	—	—
0x342		I3C_HRX	✓	✓	✓	✓	—	—
0x343		I3C_HTX	✓	✓	✓	✓	—	—
0x344		I3C_TEND	✓	—	—	✓	—	—
0x345		I3C_EEI	✓	—	—	✓	—	—
0x346		I3C_STEV	✓	—	—	✓	—	—
0x347		I3C_MREFOVF	✓	—	—	✓	—	—
0x348		I3C_MREFCPT	✓	—	—	✓	—	—
0x349		I3C_AMEV	✓	—	—	✓	—	—
0x34A		I3C_WU	✓	—	—	✓	✓	—

Table 14.5 Event List (13 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x34B	ADC16H	ADC_LIMCLPI	✓	—	—	✓	—	—
0x34C		ADC_FIFOOVF	✓	—	—	✓	—	—
0x34D		ADC_ADI0	✓	✓	✓	✓	—	—
0x34E		ADC_ADI1	✓	✓	✓	✓	—	—
0x34F		ADC_ADI2	✓	✓	✓	✓	—	—
0x350		ADC_ADI3	✓	✓	✓	✓	—	—
0x351		ADC_ADI4	✓	✓	✓	✓	—	—
0x352		ADC_ADI5	✓	✓	✓	✓	—	—
0x353		ADC_ADI6	✓	✓	✓	✓	—	—
0x354		ADC_ADI7	✓	✓	✓	✓	—	—
0x355		ADC_ADI8	✓	✓	✓	✓	—	—
0x356		ADC_FIFOREQ0	✓	✓	✓	✓	—	—
0x357		ADC_FIFOREQ1	✓	✓	✓	✓	—	—
0x358		ADC_FIFOREQ2	✓	✓	✓	✓	—	—
0x359		ADC_FIFOREQ3	✓	✓	✓	✓	—	—
0x35A		ADC_FIFOREQ4	✓	✓	✓	✓	—	—
0x35B		ADC_FIFOREQ5	✓	✓	✓	✓	—	—
0x35C		ADC_FIFOREQ6	✓	✓	✓	✓	—	—
0x35D		ADC_FIFOREQ7	✓	✓	✓	✓	—	—
0x35E		ADC_FIFOREQ8	✓	✓	✓	✓	—	—
0x35F		ADC_CMPI0	✓	—	—	✓	—	—
0x360		ADC_CMPI1	✓	—	—	✓	—	—
0x361		ADC_CCMPM0	✓	✓	✓	✓	—	—
0x362		ADC_CCMPUM0	✓	✓	✓	✓	—	—
0x363		ADC_ERR0	✓	—	—	✓	—	—
0x364		ADC_RESOVF0	✓	—	—	✓	—	—
0x366		ADC_CALEND0	✓	—	—	✓	—	—
0x367		ADC_CMPI2	✓	—	—	✓	—	—
0x368		ADC_CMPI3	✓	—	—	✓	—	—
0x369		ADC_CCMPM1	✓	✓	✓	✓	—	—
0x36A		ADC_CCMPUM1	✓	✓	✓	✓	—	—
0x36B		ADC_ERR1	✓	—	—	✓	—	—
0x36C		ADC_RESOVF1	✓	—	—	✓	—	—
0x36E		ADC_CALEND1	✓	—	—	✓	—	—
0x36F	DOC	DOC_DOPCI	✓	—	—	✓	—	—
0x371	RSIP-E50D	RSIP_TADI	✓	—	—	✓	—	—
0x382	GLCDC	GLCDC_VPOS	✓	—	—	✓	—	—
0x383		GLCDC_L1UNDF	✓	—	—	✓	—	—
0x384		GLCDC_L2UNDF	✓	—	—	✓	—	—

Table 14.5 Event List (14 of 14)

Event number	Interrupt request source	Name	IELSRn		DELSRn	Canceling CPU Deep Sleep	Canceling Software Standby	Canceling Deep Software Standby
			Connect to NVIC	Invoke DTC	Invoke DMAC			
0x385	DRW	DRW_IRQ	✓	—	—	✓	—	—
0x388	MIPI DSI	DSI_SEQ0	✓	—	—	✓	—	—
0x389		DSI_SEQ1	✓	—	—	✓	—	—
0x38A		DSI_VIN1	✓	—	—	✓	—	—
0x38B		DSI_RCV	✓	—	—	✓	—	—
0x38C		DSI_FERR	✓	—	—	✓	—	—
0x38D		DSI_PPI	✓	—	—	✓	—	—
0x38F	MIPI CSI	CSI_RX	✓	—	—	✓	—	—
0x390		CSI_DL	✓	—	—	✓	—	—
0x391		CSI_VC	✓	—	—	✓	—	—
0x392		CSI_PM	✓	—	—	✓	—	—
0x393		CSI_GST	✓	—	—	✓	—	—
0x394		CSI_DBG	✓	—	—	✓	—	—
0x395	VIN	VIN_IRQ	✓	—	—	✓	—	—
0x396		VIN_ERR	✓	—	—	✓	—	—
0x397	CEU	CEU_CEUI	✓	—	—	✓	—	—

Note 1. Only supported when CMPCTL0.CSTEN = 1.
Note 2. Only the first edge detection is valid.
Note 3. Support only interrupt after DTC transfer.
Note 4. Only supported in Deep Software Standby mode 1 and Deep Software Standby mode 2.
Note 5. Only supported in Deep Software Standby mode 1.

14.4 Maskable Interrupt Operation

The ICU performs the following functions:

- Detecting interrupts
- Enabling and disabling interrupts
- Selecting interrupt request destinations such as CPU interrupt, DTC activation.

14.4.1 Interrupt Detection Selection

The interrupt controller unit selects an event source input from a peripheral function interrupt or an external terminal interrupt with IELSRn.IELS [9:0] (n = 0 to 95).

The accepted interrupt source sets the IELSRn. IR (n = 0 to 95) to 1 and sends an interrupt request to the NVIC.

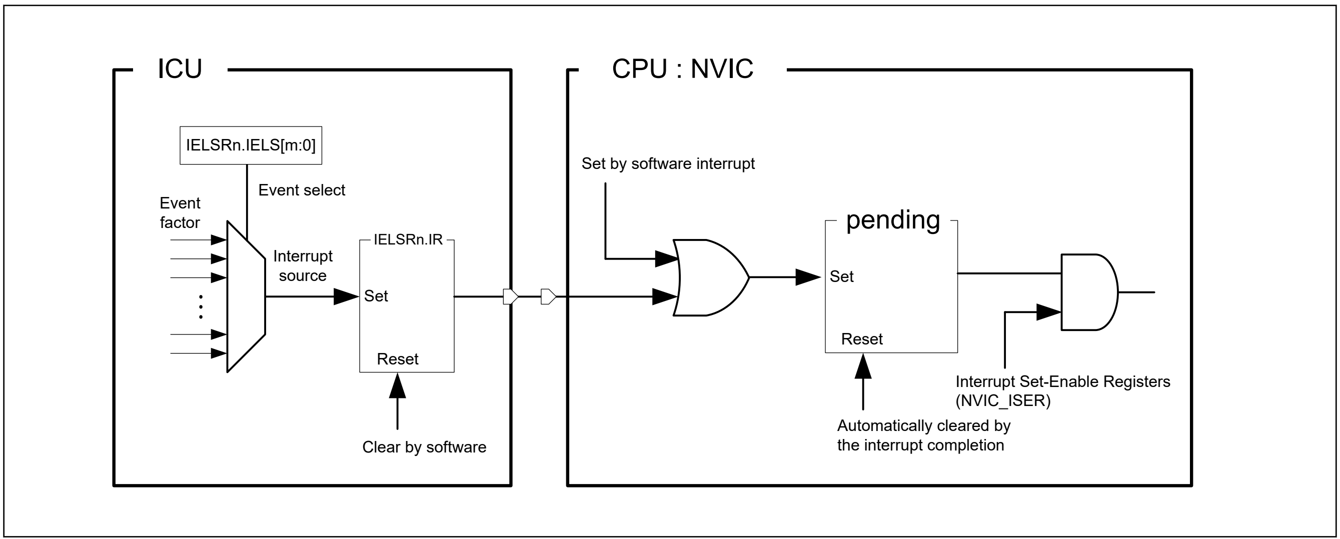

Figure 14.3 Interrupt path of the ICU and CPU : NVIC

14.4.2 Detecting Interrupts

External pin interrupt requests are detected by either the edge or level (falling edge, rising edge, rising and falling edges, or low level) of the interrupt signal. Set the IRQCRi.IRQMD[1:0] (i = 0 to 31) bits to select detection mode for the IRQi (i = 0 to 31) pins. For interrupt sources associated with peripheral modules, see section 14.3.2. Event Number.

14.5 Maskable Interrupt Setting Procedure

14.5.1 Operations During an Interrupt

- When an interrupt is generated: except for software interrupts:
  The IELSRn.IR (n = 0 to 95) flag and Interrupt Set Pending Register (NVIC_ISPRn) are set.
- When an interrupt is generated: software interrupts:
  Interrupt Set Pending Register (NVIC_ISPRn) are set.
- When an interrupt is complete:
  Clear the IELSRn.IR (n = 0 to 95) flag by software. The Interrupt Set Pending Register (NVIC_ISPRn) is automatically cleared.

Note: There may be a difference in processing speed between the CPU and ICU, and the CPU may exit the interrupt handler before clearing the IR. Then the CPU will accidentally jump to the interrupt handler again. To avoid this, be sure to read IELSRn (n = 0 to 95) register before exiting the interrupt handler and make sure that IR is cleared before exiting the interrupt handler.

14.5.2 Enabling Interrupt Requests

The procedure for enabling an interrupt request is as follows:

1. Set the Interrupt Set-Enable register (NVIC_ISER).
2. Set the IELSRn.IELS[9:0] (n = 0 to 95) bits as the interrupt source.
3. Specify the operation settings for the event source, such as Deep Sleep/Software Standby mode cancellation (WUPEN0, WUPEN1, DSLPWUPIRQENj (j = 0 to 2) register setting).

14.5.3 Disabling Interrupt Requests

The procedure to disable the interrupt request is as follows:

1. Disable the operation settings for the event source, such as Deep Sleep/Software Standby mode cancellation (WUPEN0, WUPEN1, DSLPWUPIRQENj (j = 0 to 2) register setting).
2. Clear the interrupt source setting (IELSRn.IELS[9:0] (n = 0 to 95) = 0x00).

3. Clear the interrupt status flag (IELSRn.IR (n = 0 to 95) = 0).
4. Clear the interrupt Clear-Enable register (NVIC_ICER) and interrupt Clear-Pending register (NVIC_ICPR).

14.5.4 Polling for interrupts

The procedure for polling for interrupt requests is as follows:

1. Set the Interrupt Clear-Enable register (NVIC_ICER).
2. Set the IELSRn.IELS[9:0] (n = 0 to 95) bits as the interrupt source.
3. Poll the interrupt Set-Pending register (NVIC_ISPR).

14.5.5 Selecting Interrupt Request Destinations

The interrupt output destination, CPU and DTC can be independently selected for each interrupt source.

The available destinations are fixed for each interrupt, as described in section 14.3.2. Event Number.

Use an interrupt request destination setting that is indicated by a "✓" in the event list (see section 14.3.2. Event Number).

If the CPU or DTC in one IELSRn (n = 0 to 95) register is selected, setting the same interrupt source in any other IELSRn (n = 0 to 95) register is prohibited.

Setting the same interrupt source for IELSRn (n = 0 to 95) and DELSRm is prohibited.

If the DMAC or DTC is selected as the destination for requests from an IRQi (i = 0 to 31) pin, you must set the IRQCRi.IRQMD[1:0] (i = 0 to 31) bits for that interrupt to select edge detection.

14.5.5.1 CPU Interrupt Request

When IELSRn.DTCE (n = 0 to 95) = 0, the event specified in the IELSRn (n = 0 to 95) register is output to the NVIC.

14.5.5.2 DTC Activation

When IELSRn.DTCE (n = 0 to 95) = 1, the event specified in the IELSRn register is output to the DTC. Use the following procedure:

1. Set the IELSRn.IELS[9:0] (n = 0 to 95) bits to the target event and set the IELSRn.DTCE (n = 0 to 95) bit to 1.
2. Set the DTC Module Start bit (DTCST.DTCST) to 1.

An interrupt is generated by the same factor after DTC transmission is complete.

Table 14.6 shows operation when the DTC is the interrupt request destination.

Table 14.6 Operation at DTC activation

Interrupt request destination	DISEL*1	Remaining transfer operations	Operation per request	IR*2	DTCE after transfer
DTC*3	1	≠ 0	DTC transfer → CPU interrupt	Cleared on interrupt acceptance by the CPU	IELSRn.DTCE = 1
		= 0	DTC transfer → CPU interrupt	Cleared on interrupt acceptance by the CPU	IELSRn.DTCE = 0 (IELSRn.DTCE bit is automatically cleared
	0	≠ 0	DTC transfer	Cleared at the start of DTC data transfer after reading DTC transfer data	IELSRn.DTCE = 1
		= 0	DTC transfer → CPU interrupt	Cleared on interrupt acceptance by the CPU	IELSRn.DTCE = 0 (IELSRn.DTCE bit is automatically cleared)

Note 1. DTC.MRB.DISEL bit controls the interrupt generates timing from DTC to CPU.
Note 2. When the IELSRn.IR (n = 0 to 95) flag is 1, an interrupt request (DTC activation request) that occurs again is ignored.
Note 3. For chain transfers, DTC transfer continues until the last chain transfer ends. The DISEL bit state and the remaining transfer count determine whether a CPU interrupt occurs, the IELSRn.IR (n = 0 to 95) flag clear timing, and the interrupt request destination after transfer. See Table 18.2 in section 18, Data Transfer Controller (DTC).

Note: Error during DTC transfer

If an error response occurs during DTC transfer, the DTC notifies the ICU that an error has occurred. ICU clears all bits of the target IELSRn (n = 0 to 95). IELSRn (n = 0 to 95) that is not the target is not cleared.

14.5.5.3 DMAC Activation

Events specified in the DELSRm (m= 0 to 7) registers are output to the DMAC.

To set the interrupt source for DMAC, use the following procedure:

1. Set the DELSRm.DELS[9:0] (m= 0 to 7) bits to the event to activate the DMAC.
2. When using interrupts to CPU, set the IELSRn.IELS (n = 0 to 95) bit to factor of DMAC interrupt and IELSRn.DTCE (n = 0 to 95) bit to 0.
3. Set the activation source for the target DMAC channel (DMACm.DMTMD.DCTG[1:0] (m= 0 to 7)) to 01b (interrupt module detection).
4. Set the DMAC transfer enable bit for the target DMAC channel (DMACm.DMCNT.DTE (m= 0 to 7)) to 1.
5. Set the DMAC operation enable bit (DMAST.DMST) to 1.

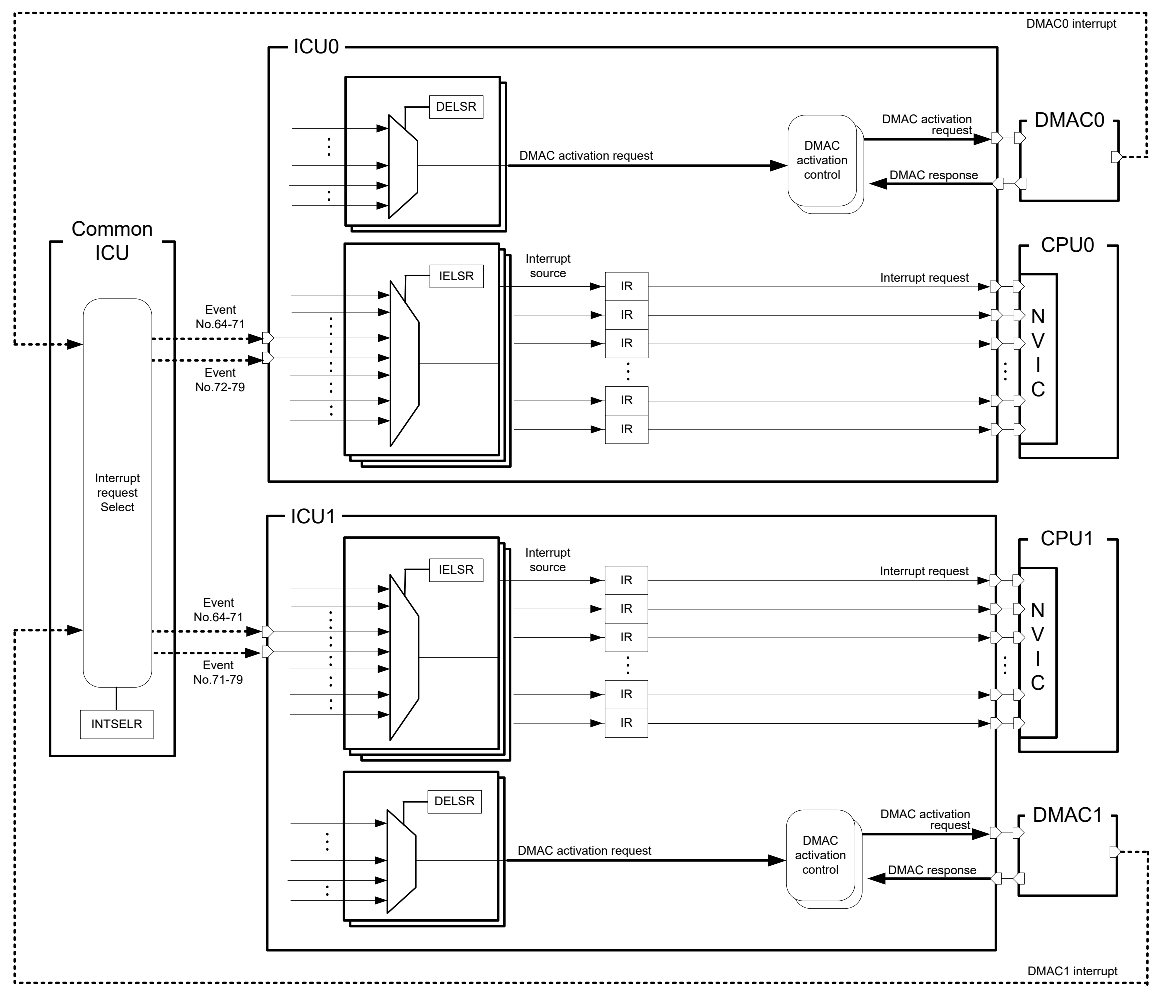

Figure 14.4 Course of the DMAC request trigger and interrupt

Note: If an error response occurs during DMAC transfer, the DMAC notifies the ICU that an error has occurred.

The ICU clears all bits of the target channel of DELSRm (m = 0 to 7). DELSRm (m = 0 to 7) that is not the target channel is not cleared.

14.5.6 Digital Filter

A digital filter function is provided for the external interrupt request pins IRQi, (i = 0 to 31) and the NMI pin interrupt. It samples input signals on the filter PCLKB sampling clock, and removes any signal with a pulse width less than 3 sampling cycles.

To use the digital filter for an IRQi pin:

1. Set the sampling clock cycle to PCLKB, PCLKB/8, PCLKB/32, or PCLKB/64 in the IRQCRi.FCLKSEL[1:0] bits (i = 0 to 31).
2. Set the IRQCRi.FLTEN bit (i = 0 to 31) to 1 (digital filter enabled).

To use the digital filter for an NMI pin:

1. Set the sampling clock cycle to PCLKB, PCLKB/8, PCLKB/32, or PCLKB/64 in the NMICR.NFCLKSEL[1:0] bits.
2. Set the NMICR.NFLTEN bit to 1 (digital filter enabled).

Figure 14.5 shows an example of digital filter operation.

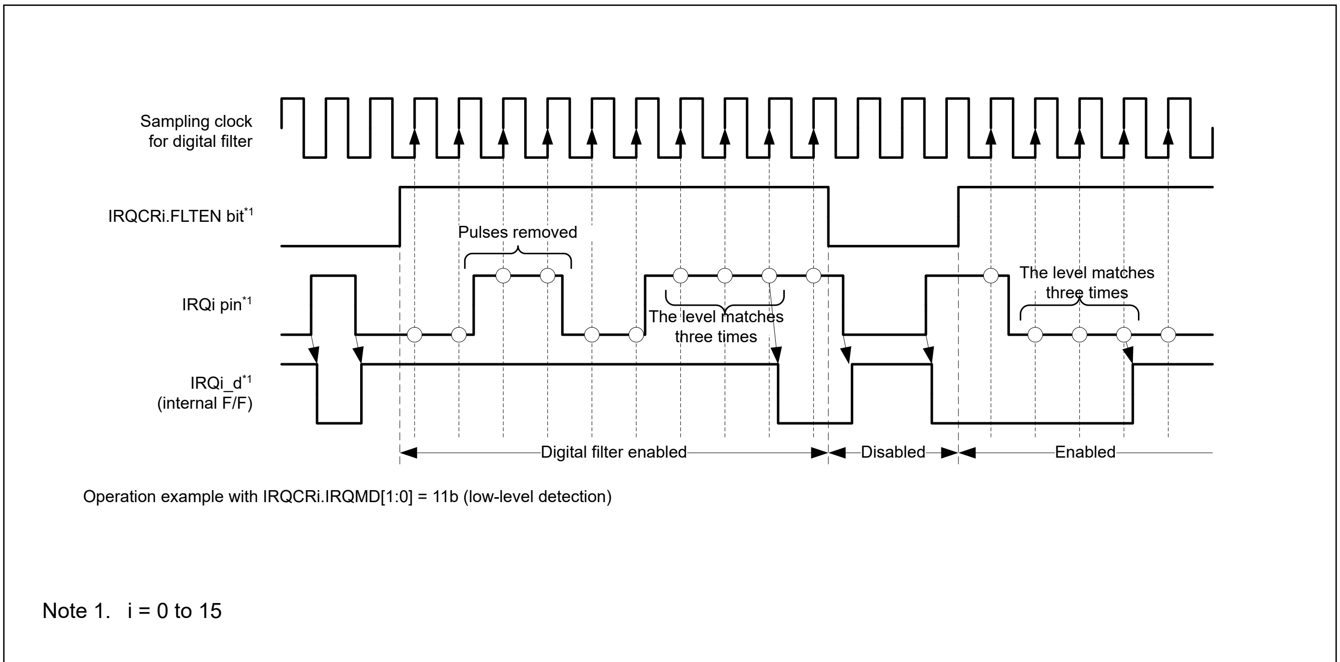

Figure 14.5 Digital filter operation example

During Software Standby mode, the digital filter is forcibly turned off by Hardware. After returning to normal mode from Software Standby mode, follow the values of IRQCRi.FLTEN (i = 0 to 31) and NMICR.NFLTEN. Once the filter is disabled, the event information sampled up to that point will be lost.

14.5.7 External Pin Interrupts

To use external pin interrupts:Configure I/O ports settings.

1. Clear the IRQCRi.FLTEN bit (i = 0 to 31) to 0 (digital filter disabled).
2. Specify or confirm the I/O port settings.
3. Set the IRQMD[1:0] the FCLKSEL[1:0] bits, and the FLTEN bit of the IRQCRi register (i = 0 to 31) register.
4. Select the IRQ pin as follows:
   - If the IRQ pin is to be used for CPU interrupt requests, set the IELSRn.IELS[9:0] (n = 0 to 95) bits and the IELSRn.DTCE (n = 0 to 95) bit to 0.
   - If the IRQ pin is to be used for DTC activation, set the IELSRn.IELS[9:0] (n = 0 to 95) bits and the IELSRn.DTCE (n = 0 to 95) bit to 1.
   - If the IRQ pin is to be used for DMAC activation, set the DMAC. DELSRm.DELS[9:0] (m = 0 to 7) bits.

14.6 Non-Maskable Interrupt Operation

The following sources can trigger a non-maskable interrupt:

- NMI pin interrupt
- Oscillation stop detection interrupt
- Sub Oscillation stop detection interrupt
- WDT underflow/refresh error interrupt (CPU0, CPU1)
- IWDT underflow/refresh error interrupt
- Voltage monitor 1 interrupt
- Voltage monitor 2 interrupt
- Common memory error interrupt
- Local memory error interrupt (CPU1 only)
- Bus error interrupt
- Lockup error interrupt (CPU0, CPU1)
- IPC NMI CPU mutual interrupt (CPU0, CPU1)
- FPU Exception interrupt (CPU0, CPU1)
- MRAM MRC/MRE read error interrupt

Non-maskable interrupts can only be used with the CPU, not to activate the DTC. Non-maskable interrupts take precedence over all other interrupts. The non-maskable interrupt states can be verified in the Non-Maskable Interrupt Status Register (NMISR). Confirm that all bits in the NMISR are 0 before returning from the NMI handler.

There may be a difference in processing speed between the CPU and ICU, and the CPU may exit the interrupt handler before clearing the NMISR. Then the CPU will accidentally jump to the NMI handler again. To avoid this, be sure to read NMISR before exiting the NMI handler and make sure that NMISR is cleared before exiting the NMI handler.

Non-maskable interrupts are disabled by default. To use non-maskable interrupts:

1. Clear the NMICR.NFLTEN bit to 0 (digital filter disabled).
2. Set the NMIMD bit, NFCLKSEL[1:0] bits, and NFLTEN bit of NMICR register.
3. Write 1 to the NMICLR.NMICLR bit to clear the NMISR.NMIST flag to 0.
4. Enable the non-maskable interrupt by writing 1 to the associated bit in the Non-Maskable Interrupt Enable Register (NMIER).

After 1 is written to the NMIER register, subsequent write access to the NMIEN bit in NMIER is ignored. An NMI cannot be disabled when enabled, except by a reset.

14.6.1 Correspondence to TrustZone-M by NMI

The NMI security is set by AIRCR.BFHFNMINS.

Although there is only one NMI per CPU, multiple factors can be set. This section describes the procedure for mixing Secure and Non-secure programs of NMI. When doing so, the NMI-related registers of the ICU are set to Secure.

Register related to Non-Maskable Interrupt :

- NMISR
- NMIER
- NMICLR
- NMICR

Figure 14.6 shows the flow.

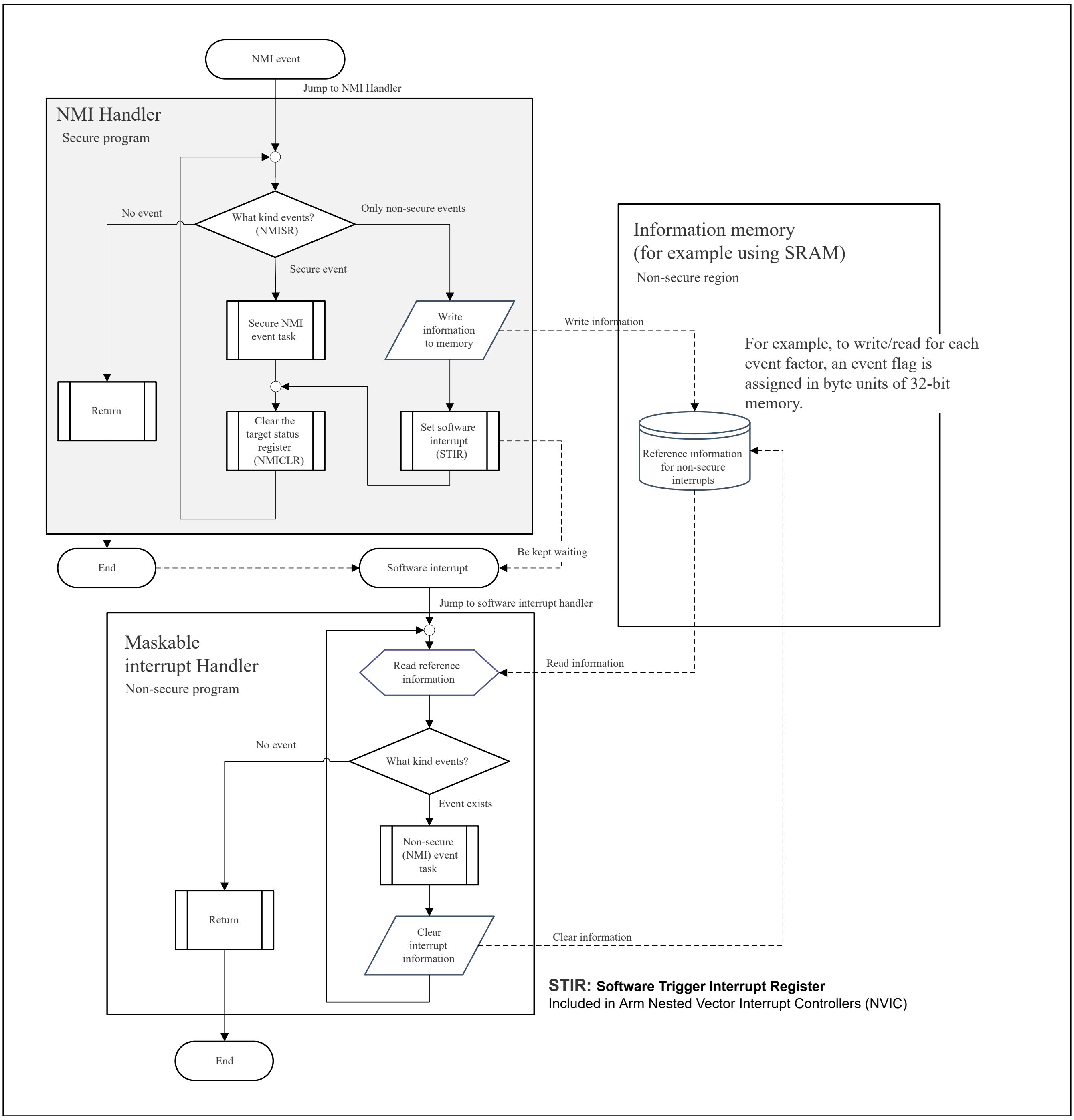

Figure 14.6 Correspondence to TrustZone-M by NMI

14.7 Inter-Processor Communication

The inter-processor communication has an interrupt function from CPU0 to CPU1 and an interrupt function from CPU1 to CPU0.

The following can be used as mutual interrupts between CPUs.

- IPC NMI CPU mutual interrupt (NMI bit20)
- IPC CPU mutual interrupt 0 (Event No.91)
- IPC CPU mutual interrupt 1 (Event No.92)

14.8 Security

14.8.1 Security Related to CPU Interrupt Inputs

Non-maskable interrupt

Arm® CPU NMI Secure is modified by AIRCR.BFHFNMINS and is managed by the software developer who manages Secure.

The following registers must follow the secure attribution set in AIRCR.BFHFNMINS.

- ICU.NMISR
- ICU.NMIER
- ICU.NMICLR

Therefore, the secure attribute set by ICUSARB must match the secure attribute set by AIRCR.BFHFNMINS.

Maskable interrupts

The secure of maskable interrupts are set in the Arm CPU NVIC internal registers (NVIC_ITNS0 to NVIC_ITNS15).

Which maskable interrupts are secured is controlled by the software developer who manages secure.

The following registers must follow the secure attribution set in NVIC_ITNS0 to NVIC_ITNS15.

- ICU.IELSRn (n = 0 to 95)

Therefore, the secure attribute set by ICUSARG, ICUSARH, and ICUSARI, ICUSARJ, ICUSARK, and ICUSARL must match the secure attribute set by NVIC_ITNS0 to NVIC_ITNS15.

14.8.2 Trusted Interrupt Management

The secure-interrupt should not be visible to non-secure-operation. Because there is a risk that vital information will be leaked to the attacker, the on-chip interrupt network can route any interrupt to secure or non-secure world. However, secure interrupt routing must be suppressed so that it can only be configured from a secure world.

When set TEVTRCR.TEVTEICU0/TEVTEICU1 = 1, secure program manages the selection of interrupt factors. It protects non-secure programs from using and monitoring secure interrupt factors without permission. Specifically, when TEVTRCR.TEVTEICU0/TEVTEICU1 = 1, the write permission of IELSR.IELS [9:0] is limited to the secure attribute, and the write by the non-secure attribute is ignored. TEVTRCR.TEVTEICU0/TEVTEICU1 protects only IELS [9:0], and the security attributes of IR and DTCE follow the settings of ICUSARG, ICUSARH, ICUSARI, ICUSARJ, ICUSARK, and ICUSARL. If TEVTRCR.TEVTEICU0/TEVTEICU1 = 1, it is necessary to set all IELS including non-secure interrupts in the secure program, or prepare a secure API to set IELS in response to the request from the non-secure program.

- Non-maskable interrupt
  The non-maskable interrupt factor that can be used by CPU0 and CPU1 is determined by NMIER register.
  The judgment conditions and priorities are as follows.
  Condition 1: If the target bit of the NMIER register is set to only one of CPU0 and CPU1, it is supplied to the set CPU.
  Condition 2: When both CPU0 and CPU1 set the target bit of the NMIER register, it is judged by the priority according to the primary/secondary of the CPU and the security level at the time of writing the NMIER register.
  Priority (1) If the NMIER register on the primary CPU side is set with a secure write, the primary CPU has priority.
  Priority (2) If the NMIER register on the secondary CPU side is set with a secure write, the secondary CPU has priority.
  Priority (3) If the NMIER register is set with a non-secure write for both CPUs, the primary CPU has priority.
  Example 1: When ICU0.NMIER.NMIEN=0, ICU1.NMIER.NMIEN= 1, supply to CPU1.
  Example 2: When CPU0 is the primary CPU.
  - When both CPU0/CPU1 set the NMIER register with a secure write, it is supplied to CPU0.
  - When NMIER of CPU0 is set by non-secure write and NMIER of CPU1 is set by secure write, it is supplied to CPU1.
  - When the NMIER register is set with a non-secure write for both CPU0/CPU1, it is supplied to CPU0.

  The exception conditions are shown below.
  Exception 1) Fixed to the primary CPU side, not selectable on the secondary CPU side.
  - bit0. IWDT underflow/refresh error interrupt

Exception 2) There is no distribution control between CPU0 and CPU1 due to individual CPU factors.

- bit1. WDT Underflow/Refresh Error Interrupt (Accepts for same CPUs).
- bit15. Lockup error interrupt (Accepts for same CPUs).
- bit16. FPU Exception interrupt (Accepts for same CPUs).
- bit20. IPC NMI CPU mutual interrupt (Controlled by IPC).

- Maskable interrupts
  The maskable interrupt factor that can be used by CPU0 and CPU1 is determined by INTSELRp register. (p = 0 to 31).
  The bit number of the INTSELRp register corresponds to the number of EventList.
  Setting the INTSELRp register to 0 will allow it to be used by CPU0, and setting 1 will allow it to be used by CPU1.
  However, the following factors can be used regardless of the setting value of the INTSELRp register.
  - Event No.88 DBG CTI0 interrupt (Accepts for same CPUs).
  - Event No.89 DBG CTI1 interrupt (Accepts for same CPUs).
  - Event No.91 IPC CPU mutual interrupt 0 (Controlled by IPC).
  - Event No.92 IPC CPU mutual interrupt 1 (Controlled by IPC).
  - Event No.101 FPU Exception interrupt (Accepts for same CPUs).

14.8.3 Trusted IELSR Setting Procedure

14.8.3.1 Case Where Secure Program Sets all IELSRs

Case where all settings are performed in the secure initial sequence.

Initial IELSR setting procedure after system reset release

1. Secure program sets interrupt request select (ICU.INTSELRp)
2. Secure program sets TEVTEICU0/TEVTEICU1 = 1
3. Secure program sets security attributes of all interrupts (NVIC_ITNS, Security Attribution of ICU.IELSRn.IELS)
4. Secure program selects all interrupt source (ICU.IELSRn.IELS)
5. Secure program allows secure interrupts (NVIC_ISER for secure interrupt)
6. Secure program jumps to non-secure program
7. Non-secure program allows non-secure interrupts (NVIC_ISER for non-secure interrupt)

14.8.3.2 Case Where Non-secure Program Sets IELSRn by Secure API

Case of setting upon receiving a request from a non-secure program.

Initial IELSR setting procedure after system reset release

1. Secure program sets interrupt request select (ICU.INTSELRp)
2. Secure program sets TEVTEICU0/TEVTEICU1 = 1
3. Secure program sets security attributes of all interrupts (NVIC_ITNS, Security Attribution of ICU.IELSRn.IELS)
4. Secure program selects secure interrupt source (ICU.IELSRn.IELS)
5. Secure program enables secure interrupt (NVIC_ISER for secure interrupt)
6. Secure program jumps to non-secure program
7. Non-secure program calls secure program (API call)
8. Secure program selects the cause of non-secure interrupt (ICU.IELSRn.IELS)
9. Secure program returns to non-secure program
10. Non-secure program allows non-secure interrupt (NVIC_ISER for non-secure interrupt)

14.8.3.3 IELSR Release Procedure When TEVTEICU0/TEVTEICU1 = 1

< When clearing interrupt settings of secure attribute by the secure program >

1. Clear the interrupt source setting (IELSRn.IELS = 0x00)
2. Clear the interrupt status flag (ICU.IELSRn.IR = 0)
3. Clear the Interrupt Clear Enable register (NVIC_ICPR)

<When clearing interrupt settings of non-secure attribute by the secure program >

1. Clear the interrupt source setting (IELSRn.IELS = 0x00)
2. Clear the interrupt status flag by using the non-secure alias address (IELSRn.IR = 0)
3. Clear the Interrupt Clear Enable register (NVIC_ICPR)

Secure can also rewrite NVIV_ICPRn_NS.

< When clearing interrupt settings of secure attribute by the non-secure program >

It cannot be canceled.

< When clearing interrupt settings of non-secure attribute by non-secure program>

1. Non-secure program calls secure program (API call)
2. Clear the interrupt source setting (IELSRn.IELS = 0x00)
3. Return from secure program to non-secure program
4. Clear the interrupt status flag (IELSRn.IR = 0)
5. Clear the Interrupt Clear Enable register (NVIC_ICPR)

14.9 Return from Low Power Modes

Table 14.5 lists the interrupt sources that can be used to exit Sleep, Deep Sleep or Software Standby mode. For more information, see section 11, Low Power Mode.

For recovery from Deep Software Standby mode, see chapter “Deep Software Standby Mode”.

Note: The return factor from low power mode must have a priority that the CPU can accept.

14.9.1 Return from CPU Sleep Mode

To return from CPU Sleep mode in response to an interrupt:

non-maskable interrupt

Use the NMIER register to enable the target interrupt request.

maskable interrupt

1. Select the CPU as the interrupt request destination.
2. Enable the interrupt in the NVIC.

14.9.2 Return from CPU Deep Sleep Mode

The MCU returns from CPU Deep Sleep mode using a non-maskable interrupt or a maskable interrupt. For the non-maskable interrupt factors exiting the Deep Sleep mode, see section 14.2.14. NMIER : Non-Maskable Interrupt Enable Register. For the maskable interrupt factors exiting the Deep Sleep mode, see section section 14.2.22. INTSELRp : Interrupt Request Select Register (p = 0 to 31).

To return from CPU Deep Sleep mode in response to an interrupt:

Non-maskable interrupt

Use the NMIER register to enable the target interrupt request.

Maskable interrupt

1. Select the interrupt source that enables return from CPU Deep Sleep mode.

2. Use the DSLPWUPIRQENj (j = 0 to 2) register to enable the target interrupt request.
3. Select the CPU as the interrupt request destination.
4. Enable the interrupt in the NVIC.

Interrupt requests through the IRQi (i = 0 to 31) pins that do not satisfy these conditions are not detected while the clock is stopped in CPU Deep Sleep mode. Similarly, it cannot detect a request for a non-maskable interrupt from a request source whose clock is stopped in CPU Deep Sleep mode. Therefore, those events cannot be used as return factors. For example, the CPU cannot return from software standby mode because the following NMI events do not occur.

- Lockup error interrupt (same CPU)
- FPU Exception interrupt (same CPU)

14.9.3 Return from Software Standby Mode

MCU goes to the Software Standby mode via CPU deep sleep state.

The MCU returns from Software Standby mode using a non-maskable interrupt or a maskable interrupt. For the non-maskable interrupt factors exiting the Software Standby mode, see section section 14.2.14. NMIER : Non-Maskable Interrupt Enable Register. For the maskable interrupt factors exiting the Software Standby mode, see section section 14.2.19. WUPEN0 : Wake Up Interrupt Enable Register 0, section 14.2.20. WUPEN1 : Wake Up Interrupt Enable Register 1, and section 14.2.22. INTSELRp : Interrupt Request Select Register (p = 0 to 31).

The maskable interrupt sources to exit the software standby mode are specified by the bits in WUPEN0 and WUPEN1. Additionally, the corresponding bits in DSLPWUPIRQENj (j = 0 to 2) need to be set to “1”.

However, if the mode is switched to S/W standby mode while the return factor in CPU Deep sleep mode is detected, the re-turn factor from deep sleep may cause the S/W standby mode to return to normal mode.

No event occurs from the function that stops in Software Standby mode. Therefore, those events cannot be used as return factors. For example, the CPU cannot return from Software Standby mode because the following NMI events do not occur.

- WDT underflow/refresh error (CPU0, CPU1)
- Common memory error interrupt
- Local memory error interrupt (CPU1 only)
- Bus error interrupt
- Lockup error interrupt (CPU0, CPU1)
- IPC NMI CPU mutual interrupt (CPU0, CPU1)
- FPU Exception interrupt (CPU0, CPU1)
- MRAM MRC/MRE read error interrupt

14.10 Using the WFI Instruction with Non-Maskable Interrupts

Whenever a WFI instruction is executed, confirm that all status flags in the NMISR register are 0.

14.11 Reference

[1]ARM Limited., ARM® Cortex®-M33 Processor Technical Reference Manual (ARM 100230)

[2]ARM Limited., TrustZone® technology for ARM®v8-M Architecture (ARM 100690_0100_00_en)

[3]ARM Limited., ARM® Cortex®-M85 Processor Technical Reference Manual (ARM 101924_0000_02_en)